



SPECIAL ISSUE: 2025 Emerging Investigator Issue

Recent advances in wearable electrochemical sensors for *in situ* detection of biochemical markers

Yanli Jiao^{1,2} and Xinge Yu^{1,2*}

ABSTRACT In recent years, wearable electrochemical sensors have been widely used for biochemical analysis. These sensors, which incorporate flexible electrodes and sensitive recognition elements on a flexible substrate, facilitate the non-invasive, *in-situ*, real-time, and continuous monitoring of target biochemical molecules in biofluids while maintaining high selectivity and sensitivity. This review provides a comprehensive examination of the principles guiding the selection of core components and the recent advances in wearable electrochemical sensors for biochemical markers in recent years. Initially, we outline the essential considerations in designing wearable sensors to detect biomarkers in biofluids, including sampling techniques, material selection, design parameters, recognition elements, sensing strategies, power requirements, data processing, and sensor integration. We emphasize the improved efficacy of recognition elements, which has been significantly enhanced by biotechnology and materials science developments, facilitating selective and sensitive detection of target components within complex matrices. Concurrently, incorporating nanomaterials and conductive polymers (CPs) has markedly improved the sensing capabilities of flexible electronics. Subsequently, we investigate recent progress *in situ* detection of biochemical markers utilizing wearable electrochemical sensors that employ advanced materials, optimized mechanical structures, and various conduction mechanisms. The notable applications stemming from these technological innovations illustrate significant improvements in sensitivity, reliability, and monitoring capabilities of wearable electrochemical sensors while enhancing user comfort. Finally, we address the current challenges and future perspectives regarding implementing clinically oriented wearable electrochemical sensors for disease monitoring and personalized medicine.

Keywords: wearable sensors, electrochemical, biochemical markers, *in situ*, non-invasive

INTRODUCTION

A biomarker is a well-defined characteristic of a biological process indicating normal biological processes, pathogenic processes, or responses to exposure or intervention, including therapeutic interventions [1]. The family of biomarkers is

complicated and can be categorized into physiologic (e.g., pulse, blood pressure), molecular (biochemical markers, e.g., genes, proteins, and metabolites), histologic (e.g., tissue and cells), and radiographic (e.g., CT, MRI, or X-ray). These metrics are commonly used as indicators for early disease diagnosis and treatment. Monitoring biomarkers through analytical methods provides essential insights into the human health state at the molecular level [1–3]. Currently, biochemical assays used in clinical diagnosis rely on bulky and expensive instruments operated by trained personnel. Most of these facilities are incapable of real-time and continuous monitoring, thus limiting their use in emergency care settings and routine healthcare. The inconvenience could be increased for patients due to frequent hospital visits or pain caused by invasive sampling (e.g., blood draws and tissue biopsies). These limitations significantly hinder their effectiveness and applicability in health monitoring, disease diagnosis, treatment, and prognosis [4].

To avoid the issues associated with central-hospital facilities, point-of-care testing (POCT) is a technical trend to support healthcare management [3]. As a result, developing sensing devices showing advantages *in situ* and real-time is urgently needed [5,6]. Towards these aims, recent biomarker detection advancements have transitioned from end-point tests to portable and wearable monitoring. Recent advances in materials science, micro-nano manufacturing, flexible electronics, miniaturization technology, and the Internet of Things (IoT) have greatly facilitated interest in the development of wearable sensors, as it is becoming increasingly feasible to integrate functionalities such as real-time sensing, data acquisition, and wireless communication into a single device [7]. A range of wearable devices has been developed for detecting significant analytes, employing optical, piezoelectric, and electrochemical methods. Among these approaches, electrochemical signal transduction is particularly advantageous due to its straightforward mechanism, which facilitates the conversion of biochemical signals into electrical signals, eliminating the need for an additional conversion process, such as the transformation from photons to electrons. In addition, electrochemical transduction is predicated on processes that occur at the electrode-analyte interface, which requires only minimal sample volumes for effective detection [8]. Meanwhile, electrochemical methods enable continuous monitoring of target analytes, allowing for the real-time capture of dynamic and rapid changes and providing comprehensive

¹ Department of Biomedical Engineering, City University of Hong Kong, Hong Kong 999077, China

² Centre for Cerebro-Cardiovascular Health Engineering (COCHE), Hong Kong 999077, China

* Corresponding author (email: xingeyu@cityu.edu.hk)

health information [9,10]. Furthermore, electrochemical methods can be readily integrated into wearable devices because of their inherent simplicity, portability, low power requirements, and affordability [11–13]. Wearable electrochemical sensors have the potential to transform health monitoring and personalized medicine by enabling *in-situ* real-time, continuous, and non-invasive monitoring of human biomarkers. Recent research on these sensors has focused on detecting biomarkers in biological fluids such as cerebrospinal fluid, urine, interstitial fluid (ISF), sweat, saliva, and tears [14–16]. Firstly, researchers have developed wearable sensors for different biofluids to broaden the range of detectable analytes, challenge trace biomarkers, and explore novel biomarkers [17]. Secondly, integrating recognition elements with advanced materials has significantly enhanced selectivity and sensitivity sensing technologies, allowing for more accurate multi-detection. Thirdly, advancements in flexible printed circuit boards (FPCB), wireless communication, and self-powered wearable devices, like biofuel cells (BFC) and fully sweat-powered electronic skin, have created new opportunities for miniaturizing wearable sensor modules [18,19]. Last, by integrating data acquisition, processing, and transmission modules, along with employing various artificial intelligence (AI) and machine learning (ML) models for data analysis, these systems can generate significant health-related insights [20].

As recent reviews have underscored the importance of modern wearable sensors and their advancements in research, this review focuses on the principles of core component selection and notable progress in wearable electrochemical sensors for biochemical markers over the past five years. The second section begins with an overview of the key considerations in developing such sensors. Initially, we summarize the advantages of utilizing biofluids for wearable sensing, detailing the various analytes present in biofluids and emphasizing the methods of biofluid sampling. Subsequently, we outline the factors primarily considered in designing wearable electrochemical sensors, including materials, flexible electronics, and biocompatibility. We then

discuss the core components of wearable electrochemical sensors, including appropriate recognition elements based on the target biomarkers and electrochemical strategies for different analytes. Furthermore, regarding the electronic components, it is essential to emphasize optimizing the power supply method, wireless data transmission, signal processing, and integration. In the third section, we summarize the application of wearable electrochemical sensors across various scenarios, highlighting recent advancements in functional materials design, mechanical structure optimization, and the integration of diverse transduction mechanisms within wearable sensing platforms. The concluding section discusses the challenges associated with contemporary wearable biosensing technologies while exploring prospects, underscoring transformative pathways for advancing wearable sensing devices into sophisticated diagnostic tools for disease monitoring, personalized medicine, health surveillance, and intelligent care.

BUILDING BLOCKS OF WEARABLE ELECTROCHEMICAL SENSOR

Wearable electrochemical sensors surpass traditional instruments, emerging as practical tools for *in-situ*, real-time, and continuous monitoring of biological and chemical markers due to their high specificity and rapidness [21]. A wearable electrochemical sensor typically comprises three key components to detect the molecules of interest: a flexible substrate, a sensing module, and a signal process module (Fig. 1). The sensing module serves as the core component for detection in wearable sensors, typically consisting of a sensitive recognition element (receptor) and an electrochemical analyzer (transducer). The recognition element provides specificity to the sensor by selectively interacting with target molecules, initiating a series of events. The transduction mechanism converts electrochemical responses of molecular recognition events into measurable electrical signals [3]. Further integration of data processing and transmission components is essential for creating a multi-

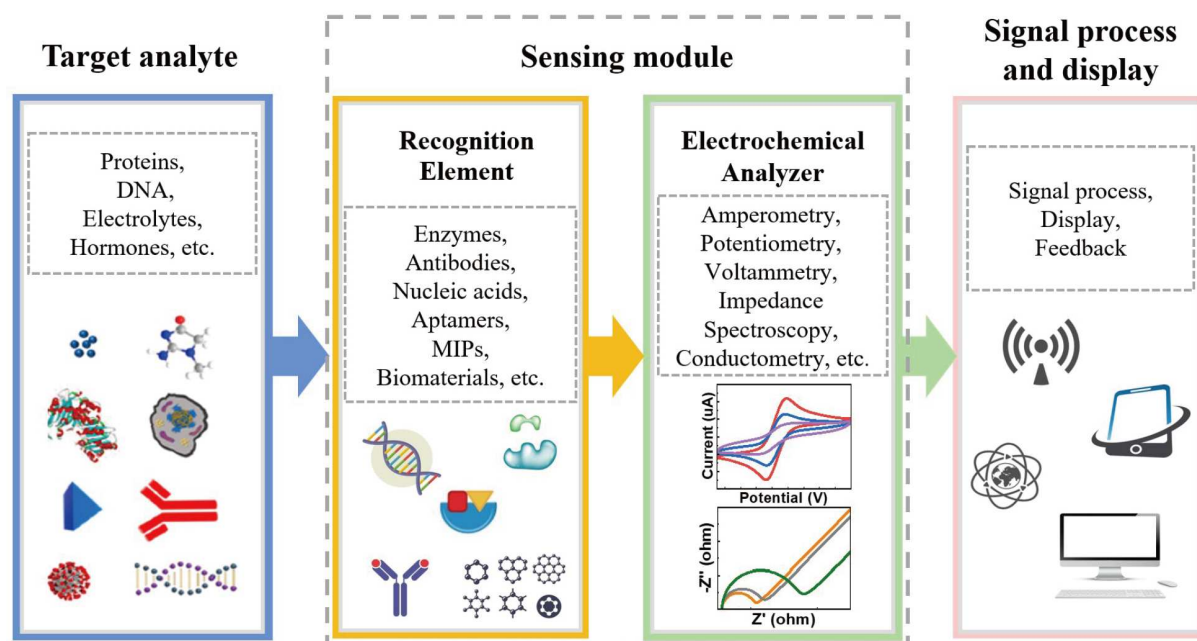


Figure 1 Main components and working principles of typical wearable electrochemical sensors.

functional platform enabling data acquisition, processing, analysis, and display [22,23].

Target analytes and sampling

Blood testing remains a standard way of assessing molecule biomarkers from the human body. However, the conventional route for blood testing requires sampling, a time-consuming, invasive, and painful procedure. Wearable electrochemical sensors capable of *in-situ*, real-time, continuous, and non-invasive detection of biomarkers in biofluids emerge as an ideal alternative to address these issues plus the sampling-induced infection risks because they can be collected and tested in a non-invasive way without damaging the outermost layer of skin or requiring blood to be drawn [24–26]. Numerous biomarkers present in biofluids, including sweat, tears, saliva, and ISF, are intricately linked to human health, encompassing electrolytes (e.g., Na^+ , K^+ , Ca^{2+} , and Cl^-), metabolites (e.g., glucose, lactate, uric acid (UA), alcohol, ascorbic acid (AA), amino acids, and ammonia), nutrients, hormones (e.g., cortisol and human chorionic gonadotropin), as well as trace amounts of macromolecules (e.g., proteins, nucleic acids, neuropeptides, and cytokines). These biomarkers have a certain degree of correlation to those in blood and can be utilized for biochemical analysis with wearable electrochemical sensors [15,22]. Consequently, it is feasible to detect various molecular markers in biofluids to monitor health, facilitate early disease detection and diagnosis, personalize treatment regimens, and develop targeted pharmaceuticals. This endeavor necessitates a comprehensive understanding of the composition of these biofluids (e.g., saliva, tears, sweat, and ISF) and the relationship between biomarker concentrations in these fluids and those in blood. Saliva-based markers are produced by salivary glands or derived from blood and easily collected, but their concentrations are lower than those in blood. When analyzing saliva for biomarker detection, it is crucial to consider potential interference from diet and the dilution effect due to saliva volume [27]. Tears also represent a valuable biofluid for biomarker analysis, while specific analyte concentrations (e.g., glucose) in tears closely correlate with those in blood [5,28]. However, continuous and sufficient sampling of tear fluid poses challenges, as it is typically obtained through various stimulation methods that yield differing analyte concentrations. Flexible, transparent, smart contact lens-based devices offer a promising platform [25]. For example, Keum *et al.* [29] have designed a wearable innovative contact lens system for noninvasive monitoring of tear glucose. Sweat is an emerging biofluid that can be collected through non-invasive stimulation methods, such as exercise, making it ideal for *in situ* monitoring via wearable electrochemical devices [30]. Although numerous biomarkers detected by wearable sensors in sweat are related to those in blood, they are typically present at much lower concentrations [5,26]. ISF is rich in blood analytes, with many metabolites present at concentration levels comparable to those in blood, as they diffuse directly through the vascular endothelium or are secreted by glands. The analysis of ISF provides more accurate sensing results than sweat, saliva, and tears, thus presenting a promising modality for health assessment [10,31]. When selecting samples for analysis, it is crucial to identify suitable biological fluids based on molecular metabolism and specific characteristics to ensure that target molecule concentrations remain detectable. Additionally, the correlation between biomarkers in these fluids and blood may be influenced

by secretion mechanisms (diffusion or leakage), secretion rates, and timing [3,32]. By comparing results with standard clinical analytical techniques, one can confirm correlations between biomarkers present in body fluids and blood, ensuring consistent trend observations.

Wearable sensors serve as an ideal platform for *in situ* biofluid analysis, where the sensing device is in direct contact with the biofluid containing the sample of interest (e.g., sweat, saliva, tears, ISF). The sample requirements of the wearable system are traced. The sampling time for tears and saliva is determined by the sampling device design, material, and capability. At the same time, sweat and ISF are more affected by how the fluid is produced and the secretion rate. In an early report, polyimide/Lycra sweat patches were used to continuously drain fresh sweat into a final reservoir for 3 h [33]. Non-invasive sampling can be accomplished through liquid guidance, surface hydrophilic/hydrophobic characteristics modifications, and incorporating highly permeable and absorbent materials and microfluidic systems to promote sampling [27,30,34]. Various absorbent materials and modified matrices such as paper, nonwovens, cellulosic, hydrogels, and rayon pads are widely used in wearable applications to minimize sampling time. These materials are characterized by their high permeability, which facilitates the extraction and collection of biofluids [35–37]. Gao *et al.* [38] placed a rayon pad between the skin and the detector to absorb and collect sweat. Based on the sweat rate ($\sim 3\text{--}4\text{ mg min}^{-1}\text{ cm}^{-2}$) and the pad size ($1.5\text{ cm}\times 2\text{ cm}\times 50\text{ }\mu\text{m}$), an optimal sampling interval of less than 1 min can be calculated to satisfy a $10\text{ }\mu\text{L}$ sample. In another study, the agarose hydrogel micropatch could be used to collect sweat samples in 1 min, and weighing the micropatch before and after 10 min of sampling showed no increase in micropatch mass [39]. Different areas, individual fitness levels, environmental conditions, and exercise intensity can significantly impact results. For example, Choi *et al.* [40] designed a network of microchannels with micro-storage chambers to gather and store sweat driven by pressure from the sweat itself. A single microstorage chamber filled after 8 min and 20 s of sweating during running. In comparison, it took 13 min and 33 s during a sauna session, corresponding to a sweat rate of $0.60\text{ }\mu\text{L min}^{-1}$ per gland from 0.8 min of sweating to a total of $1.8\text{ }\mu\text{L}$ collected over a skin area of 0.03 cm^2 . Localized stimulation techniques provide a non-invasive approach to generating adequate biofluids for wearable sensing systems. Ultrasound, iontophoresis (IP), and reverse iontophoresis (RI) can be stimulated to create sweat and ISF for efficient sample extraction [41–45]. Recently, a novel dual epidermal sampling system has been developed that utilizes cathodic chamber RI to extract ISF while simultaneously promoting sweat production in the anodic chamber through IP in the first 5 min [46]. Then, alcohol and glucose were detected by an amperometric sensor in the test samples for the following 5 min. This innovative methodology enables the simultaneous detection of alcohol in sweat and glucose in ISF. ISF is not readily accessible and requires sub-epidermal sampling, which can be done using minimally invasive microneedles (MN) [47]. For example, Mohan *et al.* [48] have developed a skin-worn electrochemical microneedle array sensor for the continuous and real-time assessment of subcutaneous alcohol in ISF. Each microneedle incorporates a microcavity that is modified alcohol oxidase (AOx), creating a stable *in situ* enzyme detection unit. Among the various techniques for sampling, hydrogel-based substrates can extract bio-

fluids in just a few minutes. Notably, hydrogels serve as a substrate for biofluid collection without high-intensity exercise or external stimulation [49,50]. For example, Sulaiman *et al.* [51] designed a microneedle array with alginate-peptide nucleic acid hybrid material to collect 6.5 μL of ISF in 2 min. However, hydrogel MNs have limitations in extracting adequate amounts of ISF, and the drive dynamics greatly influence the speed and duration of sampling. In a microneedle patch (300 μm thickness and 5 mm in diameter, featuring a 4 $\times$ 4 conical array with a height of 700 μm and each conical microneedle has a through-microchannel with a diameter of 150 μm), Rhodamine 6G (R6G) reach the filter paper at the back of the microneedle in about 26 s with hydrophilic actuation. The total extraction time was around 136 s when all microneedle channels were filled with R6G solution [52]. In another study, when ISF was extracted through a polycarbonate (PC) MN (0.2 mm in diameter) under vacuum pressure, almost no fluid entered the microfluidic chamber during the first 10 min. After about 17 min, ISF began to flow into the chamber through the microchannels in the needle, and the microfluidic chamber was nearly filled within approximately 4 min of the start of extraction [53].

Microfluidic systems developed on substrates through microfabrication techniques, laser cutting, or three-dimensional (3D) printing are highly effective in directing biofluids to detection units for *in situ* sampling and detection [27,28,35]. For example, paper microfluidic channels can facilitate the transport of sweat collected by hydrogels, with continuous evaporation from these paper channels enabling prolonged sensing using fresh sweat [50]. Numerous factors, including channel design, material choice, and fluid dynamics, can influence sampling during the design of a microfluidic channel. The operation of microfluidic devices necessitates a driving force to ensure the steady flow of bodily fluids. Besides using the biofluid as a pressure source, other mechanisms like capillary forces, osmotic pressure, and evaporation pumps can facilitate fluid movement without relying on complex external equipment. Flow direction can also be managed by incorporating valve structures or altering channel surfaces to improve hydrophilicity [7]. For instance, Yang *et al.* [54] employed a hydrogel pad in contact with the skin, integrating a polyethylene terephthalate (PET) layer as a hydrophilic microfluidic chamber (27 μL) with an additional outlet made of a paper-based sponge. The conductive hydrogel stimulated sweat secretion, filling the sensor surface within 7 min. Hu *et al.* [55] proposed a polydimethylsiloxane (PDMS)-based microfluidic system for sweat collection and detection, resulting in a flexible wearable device that enables *in situ* collection and analysis of sweat biomarkers. The filling time for the microfluidic chamber (10 μL) was estimated at 62.1 s, based on a consistent sweat production rate of about 10 $\mu\text{L min}^{-1}$. Microfluidics effectively combines various operational units, including sample preparation, reagent handling, biological reactions, and detection processes, creating a platform for miniaturization and compatibility that makes chip-based experiments feasible from conception to execution [56]. Incorporating electrochemical biosensors within the storage unit of the microfluidic device facilitates *in-situ*, real-time, and continuous monitoring of bodily fluids while also reducing sampling time and minimizing sample usage.

Material and design strategies

Recent advancements in materials science, device design, and

fabrication techniques have established a robust foundation for developing wearable electrochemical sensors, which have diverse applications throughout the human body. Flexible wearable sensors have transitioned from their original integrated forms to encompass various configurations that can be seamlessly integrated with the body, including tattoos, patches, bandages, mouthguards (MG), and contact lenses. This evolution has generated new opportunities for *in situ* monitoring of biochemical molecules through direct contact with the body. Given the necessity for conformal contact with the body while withstanding the mechanical stresses associated with movement, flexibility and stretchability are essential characteristics of wearable sensors. The design should emphasize ergonomic shapes that comfortably conform to the wearer's body, whether in a wristband, patch, or embedded within garments. The sensor must exhibit adequate durability to endure the rigors of daily activities without becoming dislodged or causing discomfort. Researchers have developed flexible wearable sensors utilizing a range of flexible materials as substrates, including flexible plastics (e.g., PET, PI, and PE), textiles, paper, hydrogels, and elastomers (e.g., PDMS and silicone rubber), to achieve seamless integration with the body's surface [11,57]. The choice of these materials allows for the development of wearable sensors that conform well to the body's contours and maintain performance over time, even under dynamic conditions. As research progresses, innovations in material science are likely to yield more sophisticated and functional wearable sensors, enhancing their application in health monitoring, sports performance, and other areas of personalized medicine.

To make wearable sensors even more flexible and deformable, there is an urgent demand for circuits and electronic components that can break free from traditional rigid configurations. Conductive materials like liquid metals, ionic liquids, and nanomaterials have been incorporated into sensor fabrication, often paired with various structural designs (such as serpentine, mesh, island-bridge, and wave-like structures) to enhance stretchability under strain. This allows smooth integration onto surfaces of varied sizes and shapes [58]. The primary materials used for creating flexible electrodes include metal nanomaterials (e.g., gold and silver nanowires and metal oxide nanoparticles), carbon-based nanomaterials (e.g., such as carbon nanotubes (CNT), and reduced graphene oxide (rGO)), as well as transition metal dichalcogenides and MXene. These materials exhibit excellent mechanical flexibility and good thermal conductivity while ensuring comfort against the skin [3]. Moreover, incorporating nanomaterials to modify electrodes can boost signal strength and detection sensitivity. Their superior electrical conductivity enhances charge transfer, improving response times. The large specific surface area and abundance of active sites can significantly expand the electrode's surface area, improve the connections between conductive components and biochemical reaction areas, and increase the availability of recognition elements and chemical reagents on the electrode surface, thereby enhancing the selectivity and sensitivity of the sensor [59,60]. Simultaneously, these nanoscale structures often exhibit impressive catalytic properties that enhance sensor performance in detecting low-concentration analytes. Conductive polymers (CPs) are known for their excellent electrical conductivity, mechanical properties, good biocompatibility, scalability, and affordability, which have led to their growing use in wearables like organic field-effect transistors (OFETs) and

organic electrochemical transistors (OECTs). However, pure CPs lack tensile strength [61]. Advanced techniques, including screen printing, transfer printing, roll-to-roll (R2R), laser engraving, deposition, inkjet printing, and 3D printing, have been employed on flexible substrates to produce highly stretchable and well-conductive films [3,62–65]. Nonetheless, their low solubility and poor melt processability pose challenges for fabrication and deposition [62]. The flexibility and stretchability of CPs can be enhanced by blending them with other stretchable polymers and incorporating metal and carbon-based nanomaterials into elastic matrices. By using these approaches, researchers can develop wearable sensors that are both highly flexible and capable of maintaining excellent electrical performance under strain. CP hydrogel-based devices are seen as ideal for designing flexible biosensors. The polymerization of CPs (like acrylamide and polyaniline (PANI)) into hydrogels (such as collagen, gelatin, alginate, cellulose, chitosan, and hyaluronic acid) results in robust and force-sensitive CP hydrogels with advantages of both CPs and hydrogel, such as high flexibility, mechanical stability, biocompatibility, excellent conductivity, hydration and water retention and responsive to various stimuli [66,67]. These characteristics make CP hydrogel-based devices highly suitable for multiple biosensing applications.

Furthermore, biocompatibility and functionality are two critical factors to consider when selecting materials for wearable sensors, given that these sensors are in direct contact with the skin [5,68]. Therefore, materials must be harmless, stable, and not irritating, causing allergic reactions or toxicity when in prolonged contact with the skin. Many biocompatible materials, including inorganic materials, such as inert metals (e.g., Pt, Au, Ti), carbon and oxide materials, organic materials including polymers (e.g., silicone materials, PET, polyester, and polyimide), small molecules, and naturally occurring biomaterials, and functionally and structurally complementary composite/hybrid materials (e.g., hydrogels), are chemically stable and can be applied to different functional layers of wearable sensors [69]. For example, the silicone elastomer PDMS is most commonly used as a flexible substrate due to its biocompatibility and high tensile strength, while AuNPs have been widely employed as sensing and conductive elements due to their low toxicity, excellent conductivity, and biocompatibility characteristics [8,17]. Additionally, permeability, breathability, and thermal management must be meticulously addressed to ensure a comfortable wearing experience, facilitate continuous monitoring over prolonged durations, and achieve stable biological signal acquisition [24,70]. Flexible electronics that utilize liquid metals and nanomaterials offer a user-friendly interface between the skin and the device. These materials can passively or actively vent and dissipate heat through their functionalized structures, ensuring a more stable signal during long-term monitoring [71].

Recognition strategies for wearable electrochemical sensor

Biomarkers in biofluids exhibit considerable diversity. When performing *in situ* measurements on the body, the pretreatment of these samples poses substantial challenges, necessitating that sensors possess the capability to identify specific target analytes from complex mixtures without interference from other components. This selectivity can be enhanced by modifying recognition elements on the electrode surface to ensure interactions with the targeted biomarker. Typically, recognition elements for biochemical markers include natural biomolecules (such as

enzymes, DNA, antibodies) and synthetic (e.g., aptamers and molecularly imprinted polymers (MIPs)) as well as ion channels and sensitive substances (e.g., CPs, electroactive materials, nanomaterials, and semiconductors) [25,72]. Recognition elements interact with target biomarkers to facilitate selective and sensitive identification and subsequently generate corresponding biochemical signals, which can be converted into readable outputs by sensing elements. Thus, high-performance recognition elements are essential for differentiating between targeted analytes and non-target components and detecting low-concentration molecules within intricate samples [24]. The selection of appropriate recognition elements depends entirely on the target analyte. Strategies employed by wearable electrochemical sensors include enzyme-based recognition, bioaffinity recognition (antibody-based recognition, nucleic acid-based recognition, and MIPs-based recognition), and ion-selective recognition. Each element possesses unique properties that facilitate the differentiation between target analytes and analogous molecular structures. This capability enhances the effectiveness of screening target analytes within complex fluids and contributes to developing more selective, sensitive, and stable wearable sensors when integrated with sensing components. Consequently, a thorough understanding of each recognition element and its influence on overall performance is crucial, offering essential insights for advancing more appropriate and robust sensors.

Enzyme-based recognition

Enzymes serve as the predominant recognition elements in wearable electrochemical sensors designed to detect a variety of analytes, including neurotransmitters, hormones, and metabolites, with a particular emphasis on glucose monitoring [73]. As biological catalysts, enzymes demonstrate significant activity, selectivity, and specificity under mild conditions, rendering them particularly suitable for sensors aimed at bio-identification within biological fluids. These enzymes are immobilized onto electrodes, where they catalyze redox reactions involving the analytes, leading to specific biological responses, such as the production of electroactive species or the facilitation of electron transfer. This process generates quantifiable signals that enable the detection of the targeted analytes [12,25]. Fig. 2a illustrates the response mechanism of various enzyme-based electrochemical sensors that measure enzyme-induced reactions, which rely on effective electron transfer between the immobilized enzyme and the electrode surface. The first generation of enzyme sensors employed oxygen as an electron carrier, enabling the quantification of either the consumption of an electroactive reactant (e.g., O_2) or the production of an electroactive product (e.g., H_2O_2) during enzyme-catalyzed redox reactions. These first-generation enzyme sensors exhibited remarkable sensitivity and were characterized by a rapid response time. Nonetheless, they faced limitations, including a high oxidation potential and susceptibility to interference from oxygen in biofluids. The second generation of enzyme-based sensors incorporates redox mediators that function as electron acceptors that directly interact with the enzyme, facilitating the transfer of electrons from the enzyme's active site to the electrode surface. The redox mediators substitute for oxygen and mitigate the oxygen effect. Ferricyanide and ferrocene are among the most widely recognized mediators. At the same time, methylene blue (MB), phenazine, methyl violet, alizarin yellow, Prussian blue (PB), thionine, azure A and C, toluidine blue, and

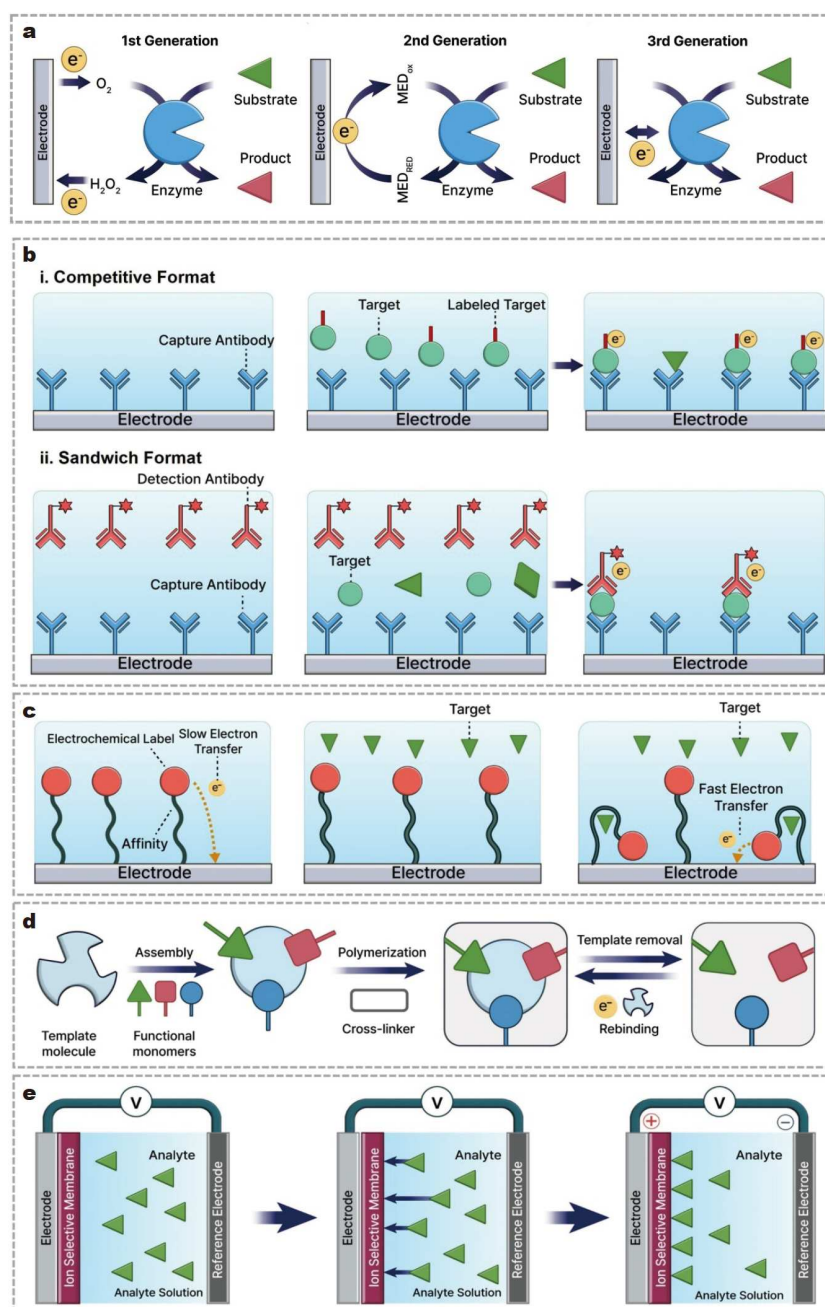


Figure 2 Response mechanism of recognition elements used in wearable electrochemical sensors. (a) Response mechanisms of 1st (electrocatalytic detection), 2nd (redox mediators), and 3rd generation enzymatic electrochemical sensors. (b) Scheme for aptamer-based electrochemical sensors of label and free-label format. (c) An electrochemical aptamer-based sensor blinds and releases a target molecule. (d) Scheme for the synthesis and re-binding of molecular imprinted polymers. (e) Scheme of an electrochemical sensor for ion measurements based on polymeric ISMs. Reprinted with permission from Ref. [12]. Copyright 2024, the Authors.

various inorganic redox ions are also commonly employed. These mediators possess a lower redox potential than other electroactive substances, which helps to reduce overpotential and minimize interference from competing redox reactions, thereby enhancing the sensors' stability, sensitivity, and selectivity [12,25]. However, solubility, biocompatibility, and leakage issues limit the practical applications of second-generation enzyme sensors. Consequently, third-generation enzyme sensors have been developed. These sensors operate without mediators by utilizing redox enzymes or engineered enzymes with modified structures to facilitate direct electron transfer (DET)

between the enzyme's active site and the electrode surface. They function within a potential window closer to the enzyme's redox potential, reducing sensitivity to interference while improving selectivity and analytical specificity [68]. In conjunction with nanomaterials, DET can be achieved using specific enzymes, such as peroxidases, pseudo-base oxidases, and fusion enzymes [74,75]. For instance, a mediator-free sensor comprising an Au/multi-walled carbon nanotubes (MWCNTs)/platinum nanoparticle (PtNPs) layer, a permselective membrane (poly-*m*-phenylenediamine, PPD) and appropriate enzyme coupled electrode demonstrates high sensitivity, selectivity, and stability

for detecting glucose, lactic acid, and choline [76]. Despite the advantages of cost-effectiveness, rapid results, and minimal sample requirements, challenges related to maintaining enzyme activity and reproducibility continue to hinder practical applications [6]. To solve this problem, target enzymes can be genetically engineered to enhance their catalytic properties and stability, as seen in recombinase applications. On the other hand, various functional materials, including polyaniline, metal oxides, and nanomaterials, can help mitigate the effects of environmental factors, thereby sustaining enzyme activity and improving immobilization efficiency to amplify enzymatic effects [31,59]. For example, incorporating nanomaterials with enzymes is increasingly utilized in complex biomolecule detection, as nanomaterials facilitate electron transfer while providing more active sites for the enzymes. The synergistic effect with enzymes enhances the catalytic current, improving the sensitivity to target analytes at low concentrations [11,43]. Notably, certain materials with electrocatalytic activity, such as transition metals (e.g., Cu, Ni, Co, and their derivatives), precious metals (e.g., Au, Pd, Ag, and Pt), various metal nanoparticles, and nanocarbon, exhibit catalytic properties akin to those of enzymes, enabling REDOX reactions without the presence of enzymes [77,78].

Antibody-based recognition

Antibodies are critical proteins within the immune system, able to accurately recognize and bind to specific target biomarkers, typically including proteins, hormones, or carbohydrates. This specificity is attributed to their unique structural features that facilitate precisely identifying epitopes and their corresponding antigen structures, rendering them significant biological recognition elements [6]. Sensors that utilize antibodies as recognition components are termed immunosensors, which are predicated on antigen-antibody interactions and are recognized for their high specificity. The interaction between antigens and antibodies can induce alterations in the surface properties of the sensor, which can be monitored through high-sensitivity detection techniques, including fluorescence labeling and electrochemical methods.

Immunosensors can be classified into two main categories: labeling techniques and label-free techniques. Labeling techniques primarily involve a competitive bind process between the target and labeled analytes at the active site of a specific antibody (Fig. 2b(i)). As the concentration of analytes in the sample increases, the binding affinity of the labeled analogs to the antibodies decreases. In contrast, label-free techniques employ two antibodies to bind with the analyte in a sandwich format, where a fixed antibody on the sensor surface acts as a capture agent, followed by a second labeled antibody that interacts with distinct epitopes on the analyte (Fig. 2b(ii)). This sandwich configuration enhances signal amplification through multiple bindings, improving the sensor's sensitivity and specificity. The label-free approach is advantageous for analyzing complex samples and quantitatively assessing numerous target analytes. At the same time, label-based immunosensors are particularly effective in detecting small molecules, even at low concentrations. Labels can encompass various materials or reagents that bind to antibodies or antigens, facilitating the evaluation of analyte quantities [12,79]. Electrochemical impedance spectroscopy (EIS) and voltammetry are commonly employed techniques for developing wearable electrochemical immunosensors to monitor antigen-antibody interactions using redox probes.

Antibodies are favored due to their availability and superior binding affinity. However, their activity can be influenced by fluctuations in temperature and pH. Furthermore, antigen-antibody recognition events are irreversible, necessitating rigorous dissociation procedures alongside microfluidic systems to refresh antibody-target complexes. As a result, recent advancements in wearable antibody sensors have predominantly concentrated on single-use designs.

Nucleic acid-based recognition

Nucleic acids, specifically DNA and RNA, serve as particular recognition elements with superior regeneration capabilities compared to antibodies. They utilize the unique complementary binding properties between DNA or RNA fragments and their corresponding target sequences to identify biological analytes accurately. However, their low stability constrains the application range of natural nucleic acid recognition elements [73,80]. In contrast, aptamers—synthetic oligonucleotide sequences composed of single-stranded DNA or RNA function as pivotal biological recognition elements in nucleic acid sensors, often called gene sensors. Aptamers are produced through the SELEX (systematic evolution of ligands by exponential enrichment) process, which involves the selection of specific sequences from a vast pool of random nucleotide sequences, yielding core sequences with variable analytical specificity alongside conserved side sequences, independent of biological systems [6,73]. Importantly, the design of aptamers facilitates the rapid generation of recognition elements that demonstrate high specificity and affinity for a diverse array of target analytes, including small molecules, proteins, peptides, nucleic acids, viruses, and cells [68]. The interactions between aptamers and analytes can occur through various physical and chemical mechanisms. Upon binding to the target analyte, the aptamer immobilized on the sensor surface experiences spatial conformational changes, which can be detected using electrochemical technologies, with or without the aid of an oxidation-reduction probe (Fig. 2c). This enhanced binding capacity is vital when structural complexity or biochemical intricacies impede antibody recognition efforts [81]. Furthermore, compared to enzymes and antibodies, the smaller size of aptamers allows for a higher density within the active layer, thereby enhancing analytical sensitivity. The stability of nucleic acids and their corresponding aptamers can be improved through chemical modifications, such as incorporating fluorescent groups or enzymes, to reduce degradation.

MIPs-based recognition

MIPs are synthetic multifunctional biomimetic receptors designed to emulate the biological recognition capabilities of antibodies, thereby functioning as “artificial antibodies”. These polymers are synthesized through molecular imprinting technology, which operates on a “lock and key” principle. As illustrated in Fig. 2d, the analyte acts as a template that facilitates the formation of customized cavities from monomers, with shapes and orientations of functional groups that correspond to those of the target molecule upon removing the template. These cavities serve as binding sites for analytes, allowing for highly selective recognition of the template molecule [6,82]. The templates can encompass a range of entities, including small molecules, ions, and larger structures such as proteins, viruses, bacteria, and entire cells [73]. Compared to other biological recognition elements, MIPs exhibit significant stability under

diverse conditions, ensuring durability and reusability while being cost-effective, as they can be produced for nearly any target analyte [11]. Although MIPs demonstrate high sensitivity towards small molecules, their sensitivity diminishes when interacting with larger molecules and biological compounds, such as proteins, viruses, and bacteria [83,84]. MIPs have been successfully utilized in wearable sweat sensors to detect non-electroactive substances, including cortisol, cardiac troponin, and cancer biomarkers [6,85,86]. As advancements in materials science, nanotechnology, and biotechnology continue, researchers are investigating novel imprinting techniques, functional monomers, and crosslinking agents to create more sophisticated and versatile MIP materials.

Ion-selective recognition

Electrolytes, including K^+ , Na^+ , Ca^{2+} , Cl^- , and NH_4^+ , are ubiquitous in biological tissues and fluids, pivotal in maintaining physiological functions and homeostasis. These electrolytes are integral to the regulation of osmotic pressure, the control of cell membrane permeability, the management of water content, and the facilitation of neural activity and muscle contraction [57,87]. Additionally, hydrogen ions are critical for sustaining the body's pH equilibrium [88]. Therefore, monitoring ionic balance in biofluids is essential for preserving healthy metabolic processes and neural functions. Ion-selective sensors are valuable for analyzing electrolytes and other biological ions, leveraging their ion-selective properties. Ion-selective membranes (ISMs) have been extensively employed in ion-selective electrodes (ISEs) and ion-selective field-effect transistors (ISFETs) for the analysis of ionic species in biofluids, which is attributed to their significant impact on the sensitivity and specificity of ion detection. The design of ISMs typically follows two primary strategies: the incorporation of specific ion receptors/ionophores or the utilization of a nanochannel or nanopore-based matrix.

Fig. 2e illustrates the dynamics of ion movement and charge separation within ISMs, generating a quantifiable surface potential. The predominant ISMs are polymeric membranes that operate on the principle of ion exchange, typically consisting of a polymer matrix, a dissolved ionophore, a plasticizer, and an ion excluder [17]. The ionophore serves as the primary active component of the ISM, governing ionic selectivity through mechanisms of electrostatic interaction or size exclusion [89]. Lipophilic salts function as ion exclusion agents, incorporating superhydrophobic elements with the same charge as the ions of interest, thereby restricting reverse ion exchange and enhancing the membrane's selectivity [90]. Plasticizers are utilized in ISMs to facilitate the dispersion of the ionophore, promote ion binding to the polymer matrix, and modulate ionic polarity, improving the flexibility and elasticity of the film and thereby optimizing the performance of the ISMs [12]. The most commonly employed plasticizers include 2-nitrophenyloctyl ether (*o*-NPOE) and bis(2-ethylhexyl) sebacate (DOS), which can be integrated with various recognition strategies [17]. Typical matrix materials encompass polyvinyl chloride (PVC), polyimide, and silicone rubber, which provide a consistent hydrophobic environment along with superior biocompatibility, thermal stability, and mechanical strength, facilitating the free movement of ionic clusters [91]. A novel approach involves the utilization of imprinted polymers on carbon substrates to create ion-imprinted polymers (IIPs), which differentiate themselves from MIPs through the interaction of the monomer or ligand

with the target ion via covalent or non-covalent bonds [92]. IIPs exhibit a high selectivity for the target ion and present a promising alternative to conventional ionophores [93].

Nanostructured ISMs signify a significant advancement in ion-selective technology, demonstrating enhanced selectivity for ions or molecules by incorporating nanopores or channels within a foundational membrane. The ion selectivity of these nanochannels is influenced by various structural attributes, including dimensions, binding sites, functional groups, charge, and hydrophilicity on their internal surfaces. Modifications to the ion selectivity of nanochannels can be achieved by altering the covalently modified surface charge through adjustments in pH, introducing chemical agents, or using photo-responsive materials. Furthermore, external stimuli, including different types of ions, electric fields, light exposure, and temperature fluctuations, can be employed to control the "on" and "off" states of the nanochannels, thereby modulating the functionality of the ISMs [94]. Due to their remarkable tunability, these membranes are attracting increasing interest and are widely applied in sensing and ion separation domains.

Sensing strategies for wearable electrochemical sensors

Electrochemical sensing in wearable sensors is a rapidly advancing field, leveraging flexible electronics and sophisticated electrode technologies to detect molecular biomarkers effectively. These sensors measure electrical signals that change in response to biochemical interactions, allowing for real-time monitoring of physiological conditions [87,95]. Electrochemical sensing is fundamentally based on the measurement of electrical signals. These methods can be categorized according to their electrical parameters into two main types: (i) potential and conductance measurements utilizing a two-electrode system and (ii) amperometric and impedance measurements utilizing a three-electrode system. Potentiometric methods are characterized as static electrochemical techniques. In the absence of external stimuli, such as applied potential or current, the accumulation of charge on the surface of the working electrode (WE) due to the selective binding of the analyte results in a stable and quantifiable potential, which can be used to ascertain the presence of the analyte. This approach has proven effective in the selective measurement of electrolytes. Furthermore, potentiometric methods have been utilized to develop immunosensors specific to proteins [13]. The amperometric technique is employed to measure the current generated between the WE and the reference electrode (RE) during the redox reactions of the analyte at a constant potential. The current produced is directly proportional to the concentration of the analyte. Various metabolites, including glucose, lactate, and alcohol, can be detected by modifying the electrode surface with a specific enzyme, thereby allowing for the measurement of the current resulting from the redox reaction between the analyte and the enzyme. In contrast, voltammetry measures the current generated by the redox reaction of the analyte as the potential between the WE and RE is varied. The concentration of the analyte is inferred from the current response to changes in potential [3]. Voltammetry generally offers superior detection limits compared to amperometry [12]. Different voltammograms can be obtained by adjusting parameters such as voltage waveform, step size, frequency, and amplitude. Differential pulse voltammetry (DPV) is particularly effective for quantitative analysis, especially in detecting trace levels of organic and inorganic sub-

stances without the need for enzymatic catalysis, as it enhances binding interactions between these substances at the sensor surface [17]. Cyclic voltammetry (CV) serves as an initial electrochemical characterization method for sensors, facilitating the exploration of electron transfer dynamics and redox processes. In contrast, square wave voltammetry (SWV) is extensively utilized for detecting trace metal elements and monitoring heavy metal contamination in the human body [23,30]. EIS is an effective technique for directly assessing electrode properties following their interaction with molecular and interfacial phenomena. This method produces impedance signals directly proportional to the analyte's activity, facilitating accurate measurement and monitoring of associated activities. In specific applications, EIS is regarded as a viable alternative to potentiometry for analyzing non-electroactive macromolecules, such as hormones and proteins [6,25]. Biochemical sensors that utilize field-effect transistors (FETs) leverage alterations in the electrical properties of semiconductors in response to local electric fields. The concentration of target biomolecules is quantified by measuring the current that flows between the source and drain electrodes. Owing to the amplification capabilities of FETs, these sensors demonstrate exceptional sensitivity and are applicable for detecting charged, neutral, and amphoteric ionic biomarkers [88,96]. Selecting the appropriate electrochemical method depends on the specific biomarker, detection sensitivity required, integration capabilities, and the environment in which the wearable sensor will operate. Further advancements in materials and technologies continue to enhance these methods, improving wearable health monitoring solutions' sensitivity, selectivity, and effectiveness.

Data processing and integration of the wearable electrochemical sensor

To enhance user comfort and convenience while minimizing disruption to daily activities, especially in environments that require continuous monitoring, wearable sensors must be compact, lightweight, and capable of providing reliable *in situ* molecular monitoring and data collection. During the design phase, these devices can be fabricated using flexible and lightweight materials, such as conductive polymers and carbon-based materials (e.g., graphene or CNTs), which help reduce size and mass while improving conductivity. Additionally, microfabrication techniques are employed to reduce the size of components, including microelectrodes and thin-film sensors. Screen-printing technology may also be considered for producing lightweight and easily manufacturable sensors. Meanwhile, integrated circuits that perform multiple functions to minimize additional parts are used. For example, printed circuit boards (PCBs) enable the incorporation of multifunctional circuits while complying with performance standards and permitting adaptable implementation. Furthermore, to assess wearable sensors, it is crucial to consider a range of factors, including power consumption, signal processing capabilities, and data transmission methods, as these aspects can profoundly influence the wearability of the devices. Minimizing the number of components and optimizing the layout to save spatial requirements is advisable. For instance, the amalgamation of sensors with supplementary functionalities, such as Bluetooth for data transmission, into a singular device can further diminish the overall weight and complexity of the system. The employment of wireless technologies, such as low-energy Bluetooth (BLE), near-

field communication (NFC), and radio-frequency identification (RFID), facilitates seamless data transfer, thereby obviating the need for extensive wiring to smartphones or laptops [25].

The development of stable and wearable power supplies is crucial for facilitating continuous and long-term *in situ* bio-sensing applications. Consequently, it is imperative to explore energy harvesting techniques, including those that harness body temperature or motion to minimize the dimensions of the power supply while simultaneously decreasing power consumption to prolong battery life. Flexible solar cells, for instance, can harness solar energy and convert it into electrical power, thereby ensuring uninterrupted system functionality [13]. Additionally, self-powered sensing strategies, which operate independently of external energy sources, utilize energy derived from human movement and biological materials to activate sensing systems, such as piezoelectric and triboelectric energy sources and biofuel cells. A prominent example of this approach is the extraction of energy from glucose and lactic acid found in biofluids. In this process, glucose oxidase (GOx) and lactate oxidase (LOx) catalyze the oxidation of these metabolites to gluconic acid and pyruvic acid, respectively, releasing electrons. Concurrently, oxygen is reduced to water at the cathode through laccase (Lac) [18,56]. Further research has demonstrated that flexible sweat-activated batteries (SABs) can generate adequate power for integrated circuits, biosensors, and NFC-based wireless micro-electronic systems, thereby enabling real-time, non-invasive sweat monitoring and addressing the limitations of self-powered systems in supporting microcontrollers [45,97]. Developing and integrating energy harvesting and small-size and lightweight storage technologies (e.g., supercapacitors) have significantly advanced the design and functionality of efficient, sustainable, and autonomous wearable systems. Furthermore, wireless energy harvesting technologies can capture energy from ambient electromagnetic fields, infrared light, and ultrasound using antennas, effectively mitigating challenges associated with continuous energy monitoring in wearable devices [3].

The integration of sensors, processors, and communication modules onto flexible circuits has the potential to yield more streamlined devices. Building signal processing functions directly within the sensor facilitates internal data analysis and interpretation, thereby minimizing the need for extensive data transmission and dependence on external hardware. This approach promotes a more efficient and compact design. Algorithms that process raw signals *in situ* and real-time prior to transmission can ensure that only pertinent information is transmitted, thereby decreasing both data volume and energy consumption [57]. The significant data output from wearable sensors necessitates integrating data analytics processes into wearable IoT platforms. The transmitted data are subsequently communicated by smartphones or laptops to a server via the Internet for application customization. For low-dimensional data, such as single or highly correlated sensor outputs, correlations between digital signals and analyte concentrations are established through transformation functions. Conversely, for complex, high-dimensional data, characterized by multiplexed and uncorrelated sensor outputs, existing data mining algorithms are inadequate to manage the anticipated data volume from wearable sensors. This challenge is addressed by developing novel algorithms and extending data mining protocols to accommodate heterogeneous information [7]. In this framework, cloud computing is pivotal for storing, processing, and

analyzing sensed data via efficient algorithms, thereby obviating the need for dedicated electronics for raw sensor data management [98]. Smart terminals, incorporating AI and machine learning algorithms, can analyze high-dimensional data and construct decision models. However, remote data processing must be accompanied by robust security measures. By prioritizing lightweight designs, energy-efficient components, integrated systems, and innovative power sources, wearable sensors' practicality and user acceptance can be significantly improved. Ongoing research and development in these domains will further advance wearable technology for health monitoring and other applications.

WEARABLE ELECTROCHEMICAL SENSORS FOR BIOCHEMICAL MARKERS MONITORING

Wearable enzymatic amperometric sensors

Enzyme-based amperometric sensors are the most widely utilized wearable electrochemical sensors. The immobilized enzyme engages with the analyte and facilitates a redox reaction, resulting in a measurable current signal directly proportional to the analyte's concentration in the sample, thereby providing an indirect assessment of the substrate amount. These enzymatic amperometric sensors demonstrate high sensitivity and specific selectivity, along with low power consumption and detection limits, making them particularly suitable for the detection of various metabolites, including glucose, lactate, UA, cholesterol, and pyruvate, as well as organic molecules such as hormones and cytokines [49,50,99–101]. These analytes are readily oxidized or reduced in the presence of enzymes. Glucose sensors constitute the most extensively researched category within this domain. These sensors can be employed on various body parts, including the skin, oral cavity, and eyes. Among these, epidermal sensors are recognized as the most convenient and unobtrusive measurement devices for users, comprising a sample collection system integrated with an instantaneous enzymatic amperometric sensor for the non-invasive monitoring of biomarkers in biofluids [63]. Hydrogels are extensively utilized as flexible substrates and biosensing materials in wearable sensors, facilitating improved adhesion, contact, and signal transmission between the skin and the sensors [102]. The incorporation of hydrogels enhances sensitivity by promoting the optimal orientation of immobilized enzymes and creating a biocompatible environment that fosters more effective detection of target analytes [103]. In a recent study, Lin *et al.* [49] developed an innovative non-invasive sweat glucose sensor (Fig. 3a) that employs hydrogels as a substrate for sweat collection, eliminating the need for high-intensity exercise or external stimulation. In this configuration, PB and poly(3,4-ethylenedioxythiophene) nanocomposites (PB-PEDOT NC) function as recognition mediators for H_2O_2 generated during the glucose redox reaction. This arrangement allows for monitoring glucose concentrations in sweat across various anatomical sites. In a separate investigation, epidermal patches incorporating multiple biosensors to detect different biomarkers in biofluids (alcohol, caffeine, lactic acid, and glucose) were developed simultaneously (Fig. 3b) [104]. IP was utilized to enhance the extraction of sweat (illustrated on the left) and ISF (illustrated on the right) via a hydrogel patch. This process involved applying stimulating agents (P^+ , nitroprusside) at the anode in conjunction with IP to optimize the sampling efficiency. In the context of enzyme-based

sensors, the WEs were modified with oxidoreductase enzymes, including LOx, GOx, or AOX, to facilitate the conversion of lactic acid, glucose, or alcohol molecules into their respective products (pyruvate, gluconic acid, or acetaldehyde), while concurrently generating hydrogen peroxide, which is utilized for current measurement. Yang *et al.* [47] reported a microneedle-based platform that integrates solid microelectrodes coated with GOx within hollow microneedles for the *in situ* measurement of glucose concentrations in ISF (Fig. 3c). CNTs combined with Nafion served as the enzyme's fixative and protective layer. In an *in vivo* study involving glucose monitoring in the ISF of rats, the system demonstrated the capability to accurately track glucose levels in healthy subjects over 14 days, showing a strong correlation with commercial glucose meters and FDA-approved continuous glucose monitor (CGM) system FreeStyle (Abbott Laboratories, Illinois, USA). Research suggests incorporating mediators such as PB, PEDOT:poly(4-styrenesulfonate) (PSS), and MB into the matrix significantly improves sensitivity and lowers detection limits [42,47,104].

In addition to epidermal sensors, various devices such as cavity sensors, pacifiers, dental retainers, and contact lenses can function as effective interfaces between the human body and electronic systems, facilitating the direct collection of saliva and tear samples for biomarker monitoring. As illustrated in Fig. 3d, a fully integrated wireless wearable platform utilizing a pacifier has been developed to detect glucose levels through a WE composed of GOx/chitosan/C-PB [105]. This device is designed to collect saliva via silicone rectifying channels and to transmit data through a compact wireless unit that incorporates an amperometric circuit and a Bluetooth communication system, thereby offering a non-invasive approach to glucose monitoring in the oral cavities of infants and newborns. Results from body sensing studies involving volunteers with type 1 diabetes indicated a strong correlation between the glucose levels measured in saliva (represented in black) using the pacifier sensor and the blood glucose concentrations (depicted in red) recorded before and after meals. In a separate investigation, a glucose sensor was developed utilizing an MG substrate composed of polyethylene terephthalate glycol (PETG) (Fig. 3e) [106]. This sensor employed a Pt WE coated with GOx/poly(2-methacryloyloxethyl phosphorylcholine (MPC)-*co*-2-ethylhexyl methacrylate (EHMA)-*co*-maltose binding protein (MBP) (PMEHB) to measure glucose levels in saliva. To reduce the impact of AA and UA on the sensor's performance, an interference-suppressing membrane made of cellulose acetate (CA) was applied to the WE. During *in vivo* evaluations, no interference with the output current was detected. Furthermore, the detection of biomarkers in tears represents a notable application of wearable electrochemical sensors. As shown in Fig. 3f, a silicone hydrogel contact lens has been developed that incorporates an enzymatic amperometric sensor, a flexible drug delivery system (f-DDS), a wireless power supply, a dedicated integrated circuit (ASIC) chip, and a remote radio frequency system for the monitoring of glucose levels and the administration of drugs via tears [29]. This system can wirelessly transmit data to control rabbits' ocular conditions. The WE of the glucose sensor consists of a Pt/Cr layer coated with polyvinyl alcohol (PVA)-chitosan hydrogel and GOx, facilitating the amperometric detection of glucose *in vitro* from tear samples. The results demonstrate a strong correlation between glucose concentrations in blood and tears in both normal and diabetic rabbit models. An alternative wearable

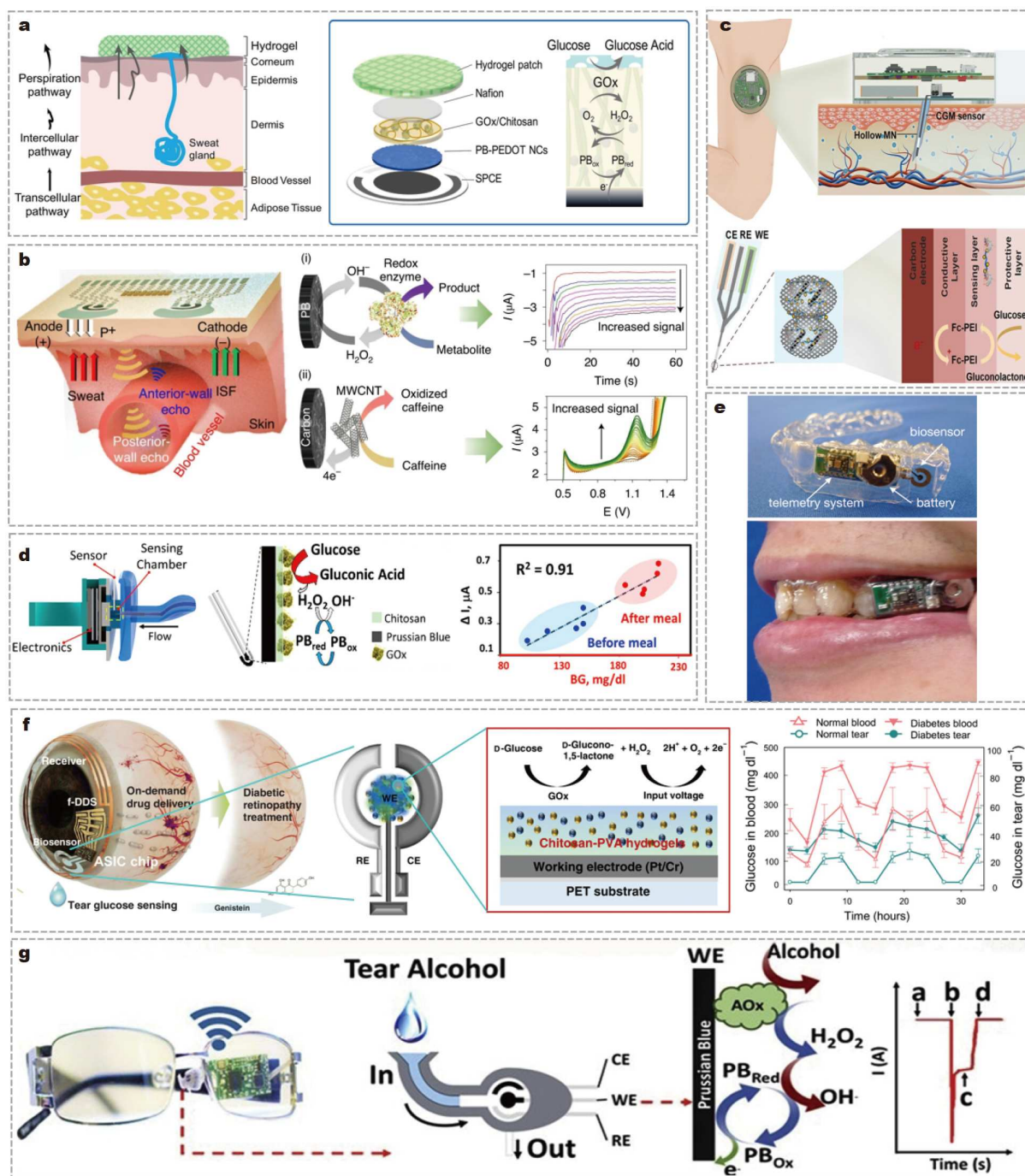


Figure 3 Wearable enzymatic amperometric sensors. (a) Schematic of the natural sweat sampling glucose sensor. Reprinted with permission from Ref. [49]. Copyright 2022, Elsevier. (b) Schematic of non-invasive, wearable patch incorporated IP with sweat-stimulating molecule (P^+ , pilocarpine nitrate) for glucose, lactate, and alcohol detection. Reprinted with permission from Ref. [104]. Copyright 2021, Springer Nature. (c) Schematic of smartphone-controlled and microneedle-based wearable CGM system. Reprinted with permission from Ref. [47]. Copyright 2023, American Chemical Society. (d) Schematic of glucose pacifier sensor with PB electrode. Reprinted with permission from Ref. [105]. Copyright 2019, American Chemical Society. (e) Illustration of a wearable cellulose acetate-coated mouthguard biosensor for *in vivo* salivary glucose measurements. Reprinted with permission from Ref. [106]. Copyright 2020, American Chemical Society. (f) Schematic diagram of a smart contact lens for diabetes diagnosis and treatment. Reprinted with permission from Ref. [29]. Copyright 2020, The American Association for the Advancement of Science. (g) Schematic of eye glasses-based biosensing system for tears glucose, vitamins, and alcohol. Reprinted with permission from Ref. [62]. Copyright 2019, Elsevier.

platform for tear sensing has been developed that is positioned externally to the ocular region, thereby mitigating the limitations associated with sensor systems that utilize contact lenses in direct contact with the eye (Fig. 3g) [62]. This non-invasive

biofluid sensing system, which employs nose pads on eyeglasses, facilitates real-time monitoring of alcohol, vitamins, and glucose concentrations in tear fluid. Chronoamperometric data collected from participants following the consumption of alcoholic bev-

erages reveal a significant correlation between the current response of alcohol in tear fluid and the blood alcohol levels measured concurrently. Additionally, a notable relationship was identified between variations in glucose levels in tear fluid and those in the bloodstream, both prior to and following meals. This continuous, non-invasive monitoring system for alcohol and glucose levels in tears has been realized by integrating biosensors into eyeglass frames. Furthermore, incorporating wireless electronic circuits within the eyeglass structure has resulted in a streamlined, portable, and user-friendly sensing device.

Wearable sensors for direct electrochemical detection

Detecting metabolites in biofluids, such as glucose, generally depends on specific enzymatic reactions. When exposed to a suitable voltage potential, these electroactive compounds can undergo electrochemical reactions (gain or lose electrons). This facilitates electron transfer at the electrode interface, leading to the generation of current or changes in potential, which is proportional to the concentration of the analyte present [11,22]. Consequently, various measurement techniques can assess the concentration and kinetic behavior of these electroactive species. Voltammetry-based sensing technologies are particularly effective in quantifying both endogenous and exogenous electroactive molecules, as they enable the direct translation of the analyte's redox properties into measurable electrical signals. Linear sweep voltammetry (LSV), DPV, and CV are frequently employed as preferred methodologies for evaluating the electrochemical behavior of these compounds at designated WE due to their high sensitivity, simplicity, and rapid operation, especially suitable for detecting low concentrations of electroactive molecules [7]. Incorporating nanomaterials and nanostructures on sensor surfaces can enhance the signal-to-noise ratio (SNR) and lower detection limits. Notably, electrocatalytically active materials, including transition metals such as Cu, Ni, and Co, along with their derivatives and precious metals like Au, Pd, Ag, and Pt, in addition to various metal and carbon nanoparticles, exhibit catalytic properties like enzymes. These materials facilitate redox reactions without enzymes and are thus utilized in non-enzymatic catalytic methods for directly detecting electroactive analytes [107]. The electrodes constructed from nanomaterials are the most frequently employed devices for the direct redox detection of electroactive compounds. Hu *et al.* [55] developed a PDMS-based microfluidic system to detect UA in sweat (Fig. 4a). The detection mechanism of the UA sensor unit is predicated on the electroactive properties of uric acid molecules, which can undergo oxidation to form allantoin, exhibiting distinct oxidation peaks in the process. The WE was modified with nano Au to increase sensitivity and lower the detected limitation, enabling the detection of UA within a concentration range of 5 to 200 μM through DPV, with a limit of detection determined to be 1.79 μM . This sophisticated technology provides significant diagnostic capabilities for routine health monitoring, particularly for individuals with gout, cardiovascular diseases, and renal disorders. Yang *et al.* [65] developed an innovative sensor that combines graphene-based chemical sensors with microfluidic systems for the continuous monitoring of low concentrations of UA and tyrosine (Tyr) in sweat (Fig. 4b). The selective oxidation of UA ($\sim 0.39\text{ V}$) and Tyr ($\sim 0.64\text{ V}$) was catalyzed at a specific potential, generating two distinct current peaks. Utilizing DPV to measure the amplitudes of the oxidation current peaks

enabled the assessment of ultra-low concentrations of UA and Tyr. The electrocatalytic activity and reproducibility sensitivity for the direct oxidation of UA and Tyr at physiological concentrations were recorded at 3.50 and 0.61 $\mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$, with low detection limits of 0.74 and 3.6 μM , respectively, demonstrating trends that were consistent with those found in serum samples. Similarly, in epidermal patches that simultaneously monitor metabolic biomarkers, carbon electrodes modified with MWCNTs can increase the electrocatalytic reaction of caffeine, thereby measuring caffeine concentration via oxidation signals [104].

The integration of various electrocatalytic and nanomaterial components not only involves the distinct properties of each material but also enhances the electrocatalytic and transduction efficiencies through synergistic interactions. For instance, 3D porous double-structured aerogels composed of gold nanowires (AuNW) and N-doped graphene nanosheets (N-rGO/Au DAs) exhibit extensive active surfaces, numerous targeted channels, rapid electron transfer pathways, and elevated intrinsic activity [108]. Consequently, the direct electrooxidation of uric acid on N-rGO/Au DAs demonstrates significantly greater activity than the individual Au and rGO components. Toi *et al.* [109] devised a non-enzymatic electrochemical patch employing textile materials for glucose monitoring in sweat (Fig. 4c). This sensor utilized a wrinkled, stretchable nanohybrid fiber (WSNF) as the electrode composed of rGO and polyurethane. Continuous glucose monitoring was facilitated by the current response of the WSNF to fluctuations in glucose concentrations. The WSNF exhibited notable electrocatalytic activity, which can be attributed to the synergistic interactions between the manifold structures and the oxygen functional groups present within the rGO matrix. This interaction was instrumental in promoting the dehydrogenation step during glucose oxidation. As a result, the WSNF electrode demonstrated an elevated level of responsivity to glucose, characterized by increased sensitivity ($140\text{ } \mu\text{A mM}^{-1} \text{ cm}^{-2}$), a low detection limit of 500 nM, significant selectivity against potential interferents, and stability across various environmental conditions. Additionally, glucose was successfully detected in neutral solution conditions. A recent investigation has demonstrated that a non-enzymatic glucose sensing chip constructed from a dual-structured Pt-Ni hydrogel exhibits remarkable electrocatalytic performance in glucose oxidation under neutral conditions. The catalytic current densities observed in Pt-Ni hydrogels were found to be 2.0 and 270.6 times greater than those recorded for pure Pt and Ni hydrogels, respectively. This chip not only displays high sensitivity and selectivity but also maintains long-term stability over two months. Consequently, noble metal aero-/hydrogels (NMG) are regarded as promising candidates for the advancement of wearable sensors owing to their superior electrocatalytic activity, favorable flexibility, and self-healing characteristics [78]. Understanding the role of direct electrochemical reactions in these sensors is fundamental to developing more sensitive and specific devices to avoid the limitations caused by enzyme-based sensors. This knowledge not only aids in designing new sensors but also in advancing their applications in various biomarker detection.

Wearable affinity electrochemical sensors

In the field of biofluids, the monitoring of trace non-electroactive macromolecules and complex metabolic analyses, such as

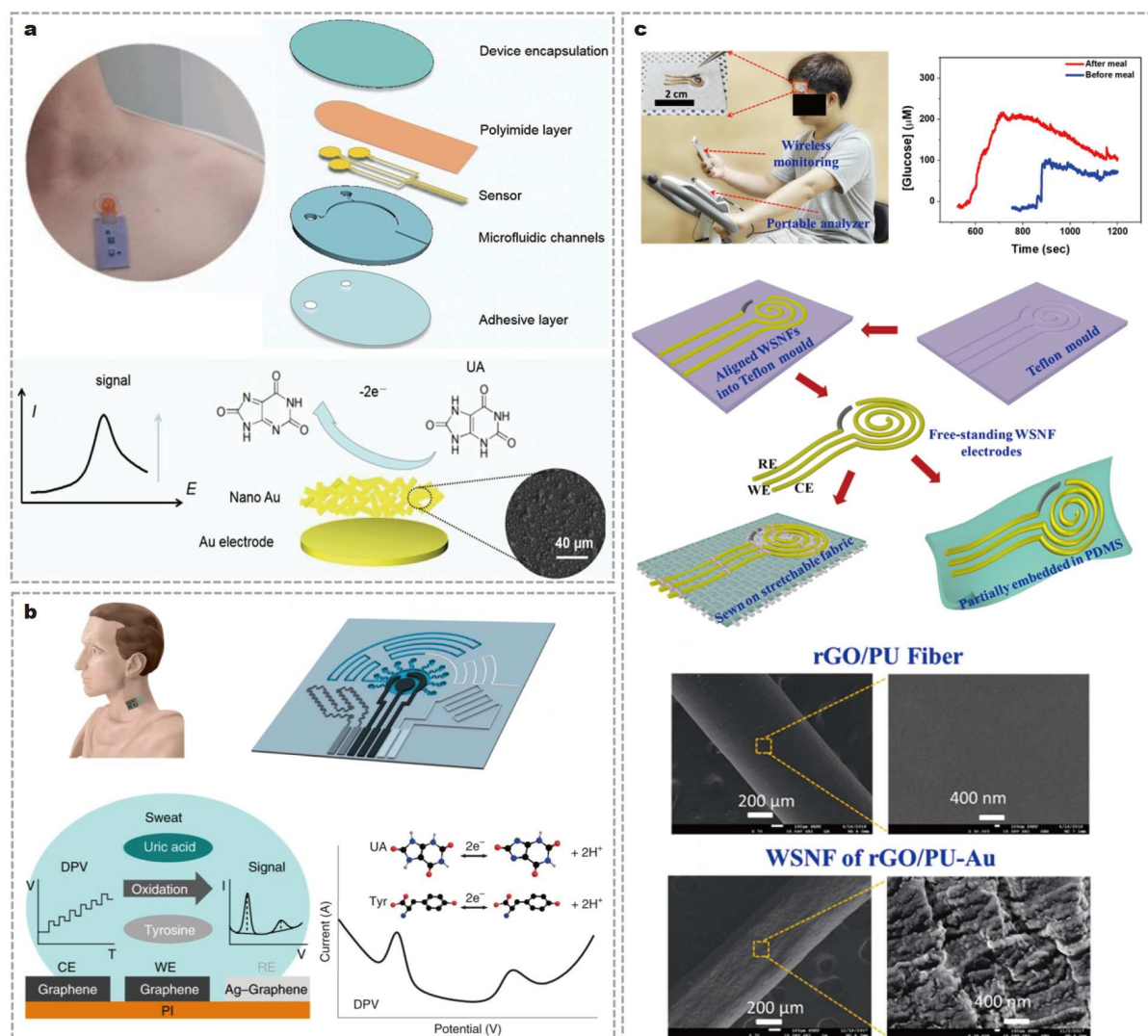


Figure 4 Wearable sensors for direct electrochemical detection. (a) Illustration of the flexible and wearable device for UA detection in sweat. Reprinted with permission from Ref. [55]. Copyright 2023, the Authors. (b) Schematics of the sweat sensor with microfluidic module for simultaneous UA and Tyr detection with DPV. Reprinted with permission from Ref. [65]. Copyright 2019, Springer Nature. (c) Architecture and fundamental characteristics of the WSNF used as an electrochemical sweat glucose sensor. Reprinted with permission from Ref. [109]. Copyright 2019, American Chemical Society.

proteins, peptides, hormones, and neurotransmitters, frequently utilizes affinity recognition techniques (e.g., antibodies, aptamers, and MIPs) to bind target molecules selectively. These affinity methods exhibit significant specificity and selectivity for various biomarkers, even at low concentrations. When integrated with electrochemical techniques, including SWV, CV, DPV, CA, EIS, and FET, wearable sensors are established as promising platforms for detection [73,96]. Antibody-based immunosensors, in particular, demonstrate high specificity for target antigens and have been employed in numerous studies to detect antigens or proteins. For instance, Jang *et al.* [110] reported a wireless monitoring and treatment system for ocular surface inflammation (OSI), which utilized a graphene FET biosensor functionalized with immunoglobulin G (IgG) to quantify matrix metalloproteinase-9 (MMP-9) concentrations in tear fluid (Fig. 5a). Additionally, they developed a stretchable, transparent heat patch that can be securely attached to the eyelid for therapeutic purposes. The system incorporates silver nanofibers to integrate wireless antennas, capacitors, resistors, and

NFC chips, utilizing smartphones for wireless data transmission. This configuration enables automatic diagnosis of the OSI and the provision of heat therapy. *In vivo* experiments have confirmed the biocompatibility and practicality of the system. Cheng *et al.* [111] demonstrated a battery-free, wireless, and flexible electrochemical patch capable of conducting an *in situ* analysis of cortisol (a glucocorticoid) in sweat. This fully printed flexible electrochemical immunosensor incorporates AuNPs that have been electrodeposited onto a screen-printed conductive ink electrode, which is subsequently functionalized with anti-cortisol antibodies to form the recognition domain. The specific interaction between cortisol and the anti-cortisol antibodies facilitates a direct correlation between the DPV detection of the conductive material and its concentration. Furthermore, integrating an NFC module allows wireless energy harvesting and data transmission to smartphones, enabling real-time testing with the patch. Recent investigations have also highlighted the advancement of immunoassay sensors for cortisol detection across various bodily fluids, employing functionalized electrodes

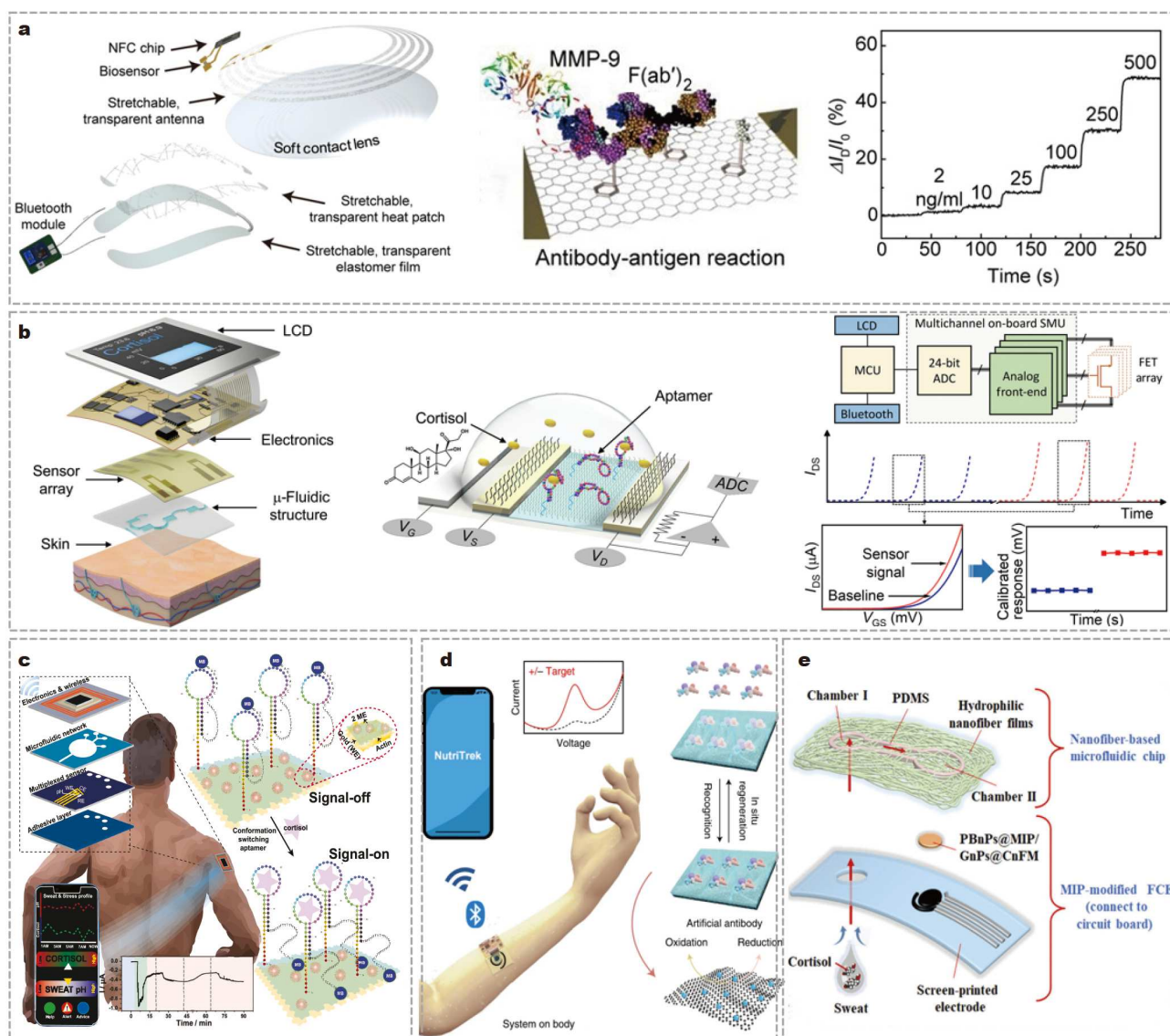


Figure 5 Wearable affinity electrochemical sensors. (a) Smart contact lens and transparent heat patch for remote monitoring and therapy of chronic ocular surface inflammation using mobiles. Reprinted with permission from Ref. [110]. Copyright 2021, The American Association for the Advancement of Science. (b) Aptamer-FET-based thin In_2O_3 sensor for the detection of sweat cortisol. Reprinted with permission from Ref. [112]. Copyright 2022, The Authors. (c) Illustration of wearable sweat patch using a pseudoknot-assisted aptamer for monitoring cortisol levels. Reprinted with permission from Ref. [113]. Copyright 2023, Elsevier. (d) Schematic of a wearable electrochemical biosensor with MIPs for monitoring metabolites and nutrients. Reprinted with permission from Ref. [114]. Copyright 2019, Springer Nature. (e) Schematic of the wearable MIPs electrochemical sensor with nanofiber-based microfluidic for cortisol analysis. Reprinted with permission from Ref. [116]. Copyright 2023, Elsevier.

composed of materials such as graphene, Ag, and metal-organic frameworks, exhibiting enhanced selectivity, sensitivity, and stability [90]. Aptamers exhibit a high degree of selectivity and affinity comparable to antibodies, demonstrating significant binding capabilities to various target molecules [73]. Their specificity for targets can be enhanced through structural modifications, improving their stability for wearable electrochemical sensors. As shown in Fig. 5b, in a flexible FET sensing system, the thiol-modified cortisol aptamer, serving as the biorecognition element, is covalently immobilized onto a nanometer-thin indium oxide (In_2O_3) channel of the FET, which acts as the signal transduction platform [112]. This system is integrated with a custom on-chip multi-channel source measurement unit (SMU) designed to detect and bind cortisol. It

converts charge variations on the surfaces of conductive materials into electrical signals on the FET, thereby facilitating cortisol detection while establishing a correlation between cortisol levels in saliva and sweat samples. In a recent study, Singh *et al.* [113] designed a microfluidic wearable sensor that employs conformation-switching aptamers for the non-invasive assessment of cortisol concentrations in sweat (Fig. 5c). The surface of the electrode was altered by incorporating a conformation-switching cortisol aptamer, which features a loop and two stems that extend into a complementary poly(T12) oligonucleotide region, thereby creating a pseudo-junction structure. This design restricts the aptamer to two distinct states, thereby optimizing the reduction of background signals and improving sensitivity. A covalently conjugated redox tag MB exhibits high specificity for

cortisol and possesses regenerative properties that facilitate continuous monitoring of temporal fluctuations. The pseudo-junction-assisted aptamer is characterized by a bistable structure. In the absence of cortisol, the probe maintains its pseudo-junction configuration, with the MB positioned away from the sensor surface. Upon the binding of cortisol, the conformation shifts from the pseudo-junction form to a monostem-loop structure, resulting in the MB's proximity to the electrode surface, which enhances electron transfer rates. This conformational change leads to current variations directly proportional to cortisol concentration. The sensitivity of this wearable electrochemical sensor has been demonstrated through voltammetric analysis, revealing a broad dynamic range (1 pM–10 μ M), an exceptional detection limit of 0.2 pM, and favorable linear relationships.

MIPs are synthetic polymers engineered with specific recognition sites. Integrating MIPs with electrochemical detection techniques holds great promise for developing sensitive, selective, and robust biosensors for various metabolites. This combination can significantly enhance analytical capabilities in clinical diagnostics, sports medicine, and metabolic studies. Considerable advancements have been achieved in the detection of various metabolites with MIPs, including amino acids [114], lactic acid [115], and urea [17], mainly through DPV. This progress highlights the efficacy of non-invasive wearable biosensors that integrate regenerated MIPs and nanomaterials (such as CNTs and AgNTs). Wang *et al.* [114] present a wearable electrochemical biosensor designed for the continuous monitoring of trace concentrations of various metabolites and nutrients in sweat, both during exercise and at rest (Fig. 5d). This biosensor is constructed with repeatedly *in situ* regenerated graphene electrodes and is functionalized with MIPs specific to metabolites and REDOX activity reporting nanoparticles. It incorporates several advanced features, including IP and microfluidics sampling, sophisticated signal processing and calibration, and wireless communication capabilities. In trials with volunteers, the biosensor successfully tracked the intake and levels of amino acids in real-time during physical activity while also evaluating the risk of metabolic syndrome by correlating amino acid concentrations in serum and sweat. The ability to monitor metabolites for the early detection of abnormal health conditions has the potential to enhance the implementation of precision nutrition strategies. In addition, MIPs demonstrate commendable selectivity even in the context of non-electroactive materials. In a study conducted by Mei *et al.* [116], a flexible electrochemical sensor was developed that incorporates a microfluidic membrane and MIP technology for the *in situ* evaluation of cortisol concentrations in sweat (Fig. 5e). A super hydrophilic porous carbon nanofibers functionalized with AuNPs (GnPs@CnFM) enabled rapid pumping of sweat into the bottom sensing chamber for collection and processing, and GnPs@CnFM served as the substrate for MIP deposition, enhancing both stability and sensitivity of the sensor. PB nanoparticles (PBnPs) were “built-in” redox probes within the MIP framework to create specific cavities that accurately capture cortisol molecules. In real-time *in vivo* testing scenarios, this sensor demonstrated exceptional accuracy across an extensive detection range from 1 nM to 1 μ M and excellent speed and reliability. Furthermore, it is connected to a flexible wireless PCB for *in situ* signal processing, enabling direct data transmission to users via Bluetooth.

Affinity-based flexible sensors are gaining recognition as a promising electrochemical platform for biological sensing, attributed to their high selectivity, low detection limits, and compact design. Nonetheless, these biosensors frequently necessitate intricate pretreatment processes and demonstrate low efficiency, coupled with elevated costs. Additionally, most affinity-based biosensors are predominantly engineered for single-use applications, which require multiple incubation steps and washing procedures to restore their receptors. Consequently, regenerating the binding interactions between analytes and bioreceptors presents significant challenges for continuous monitoring applications [23]. Future research endeavors should prioritize the optimization of analytical procedures while simultaneously advancing material innovations to improve the regeneration capabilities of these sensors.

ISE

ISEs represent the most utilized wearable potentiometric sensors. These sensors operate with a pair of electrodes and a potentiometer to measure potential variations without power consumption, thereby offering a low detection limit and a broad dynamic range of up to 8 units [117]. Owing to their intrinsic properties, which include simplicity, high sensitivity, strong selectivity, cost-effectiveness, miniaturization, ease of operation, and low energy consumption, ISEs are extensively employed in biomedical applications for the measurement of pH levels and various ion concentrations, such as Na^+ , Ca^{2+} , Mg^{2+} , and K^+ in sweat. Additionally, they are utilized to assess specific metabolites, including creatinine, glucose, urea, and some oxidation-reduction inert species [1,17,94,118].

ISE primarily comprises an ISM, a WE, an RE, and a reference solution. The ISM plays a pivotal role in determining the sensitivity and selectivity of the sensor, as its mechanism is designed to selectively bind target ions [119]. The ISM allows for the selective passage of specific ions based on their charge and size, leading to variations in charge and the generation of an electrical potential difference across the membrane. This potential difference between the membrane and the reference electrode is directly proportional to the logarithm of ion activity, thereby enabling the monitoring of ion concentration [12,30,91,120]. It is essential to choose suitable materials to construct the interface between the ISM and the surface of the WE (specifically the ion-electron transducer) to ensure optimal stability and reproducibility of the sensor's operation.

Currently, wearable ISEs employ solid-state ISEs (SC-ISEs), which replace liquid contacts with solid alternatives to address the limitations associated with liquid contacts in terms of shape and size. SC-ISEs facilitate miniaturization and optimize ion signal transduction [120,121]. These devices can be seamlessly integrated into low-power electronic systems for multifunctional *in situ* monitoring, enabling continuous real-time tracking of various cation concentrations [122–124]. Typical applications include wearable sensors, where robust solid contacts are essential for effective signal conversion and precise ion detection. These solid contacts typically consist of CPs, redox-active self-assembled monolayers, and high-capacitance nanomaterials [31]. Conducting polymers, such as polypyrrole (PPy), PEDOT: PSS, polythiophene (POT), *N,N'*-dimethyl-1,4-phenylenediamine (NNMP), and PANI, are particularly suitable due to their redox activity, which facilitates direct charge transfer between ions and electrons. These materials exhibit dual conductive

properties and convert ions into measurable electrical signals through doping-deintercalation [121]. They can be easily cast or applied via electrochemical pulse deposition onto conductive electrodes. Recent studies have shown that wearable sensors integrating CPs with SC-ISEs can effectively detect Na^+ and K^+ in human sweat while minimizing biases associated with repeated use [125,126]. Moreover, the performance of dual-layer capacitance solid contacts can be enhanced by increasing the interface area between the electrode and the ISM through drip-coated chemical deposition using nanomaterials, such as metal nanoparticles, metal oxides, carbon black, and rGO [121,127]. Recent research has demonstrated that Au NW [123,128], MWCNT [129], and graphene [130] serve as efficient ion-electron transducers within microfluidic structures, exhibiting high selectivity, repeatability, and stability for analyzing cation concentrations in sweat, including pH, Cl^- , K^+ , and Na^+ . Compared to CNTs and other conventional conducting polymers, the rapid one-step electrochemical pulse deposition process of metals and metal oxides offers significant advantages [131,132]. Furthermore, interface materials are printed between the WE and the ISM using various printing technologies, such as screen printing, inkjet printing, and aerosol spraying, to facilitate effective interactions among diverse materials. Ozer *et al.* [133] pioneered a template printing method on PET for real-time K^+ detection within wearable IoT systems. The compact PCB comprises a data processing module powered by a small battery alongside an additional module dedicated to wireless communication. This system calculates K^+ concentration and displays the results on an organic light-emitting diode (OLED) screen. A notable trend in flexible wearable SC-ISE devices is the integration of SC-ISEs with wireless technology (PCB), enabling data collection via smartphones or smartwatches while providing real-time feedback to end-users for on-site analysis of cations in sweat [122,128].

On the other hand, these materials not only function as contact layers but also serve as components of ISMs [117]. Recent studies have introduced an innovative approach that employs functionalized nanostructured ISMs, such as nano-channel membranes modified with functional groups derived from nucleic acids, antibodies, enzymes, channel proteins, or other biomolecules, utilizing the principles of MIPs applied to specialized polymers (e.g., carbon-containing PVC). This methodology effectively replaces traditional ionophores and achieves high selectivity for target ions [17]. Such membranes offer several advantages, including high stability, ease of preparation, and potential for miniaturization. MIP-based electrochemical biosensors are extensively utilized to detect drugs, inorganic ions, proteins, and nucleic acids. Zou *et al.* [134] developed a nanofluidic membrane composed of regenerated cellulose/GoO to monitor ions in sweat. This nanostructured ISM features well-aligned nanopores with diameters ranging from 3 to 4 nm. It possesses a negatively charged surface, demonstrating impressive mechanical properties, including strength (76.27 MPa), toughness (24.56 MJ m⁻³), and excellent ionic conductivity (0.65 S m⁻¹). The system exhibits remarkable sensitivity at low salt concentrations, with a low limit of detection compared to bulk electrolyte solutions.

CONCLUSIONS AND OUTLOOKS

The increasing demand for rapid and accurate *in situ* biochemical detection has generated substantial opportunities for

advancing wearable sensors. Wearable electrochemical sensors facilitate non-invasive, real-time monitoring of specific biochemical molecules within biological fluids. A notable trend in this domain is the development of miniaturized wearable sensing devices that consolidate all phases of biochemical analysis into a singular unit. Multi-sensor systems within a multi-modal framework incorporate fluid sampling capabilities, selective sensing mechanisms, compact batteries or self-powered devices, and wireless electronic components for data processing. This integration enables the extraction of multiple health-related datasets from a sole source. Techniques employing IP, RI, and microneedle in conjunction with microfluidics are expected to enhance in-site sampling and detection. The integration of recognition elements such as aptamers, MIPs, and nanoparticle channels, alongside innovative, flexible conductive materials on flexible substrates that effectively engage with skin surfaces, has significantly improved the performance of wearable electrochemical sensors. This improvement is reflected in enhanced selectivity, sensitivity, reliability, and user comfort, thereby promoting their widespread implementation in biochemical detection applications. For example, enzyme-catalyzed electrochemical biosensors necessitate further exploration regarding their stability and resilience to environmental factors. The emergence of non-enzyme-based sensors, which can directly detect target analytes in biofluids using nanomaterials independent of enzymatic reactions, offers several advantages, including increased stability, extended shelf life, and reduced costs. In addition, the implementation of efficient energy management systems has the potential to optimize energy consumption while simultaneously enhancing stability. From the perspective of data presentation, algorithms designed to process raw signals *in situ* and in real-time can ensure that only pertinent information is transmitted, thereby minimizing both data volume and energy expenditure. Wireless communication facilitates the seamless transfer of data to smartphones or laptops, enabling real-time monitoring and transmission of biomarker fluctuations through feedback mechanisms, effectively conveying health information, and providing alerts when necessary. Additionally, integrating artificial intelligence and machine learning within the IoT platform allows for the development of intelligent systems capable of processing and analyzing data, offering personalized health insights. This advancement not only improves predictive modeling related to disease progression and the assessment of therapeutic efficacy but also enhances our overall understanding of health.

Numerous innovative wearable electrochemical devices have demonstrated significant practical applications, with several systems successfully undergoing clinical trials. Nevertheless, the implementation of these devices in real-world scenarios has been hindered by a lack of widespread commercial success. Their ability to continuously monitor in-body molecular analytes is currently limited to a narrow range of metabolites and electrolytes. As a result, there is an urgent demand for the next generation of wearable electrochemical sensors that can detect a wider array of candidate biomarkers and incorporate multiple sensing modalities into a single device capable of simultaneous detection of various analytes. In this regard, developing new affinity sensors equipped with suitable signal labels, *in situ* non-invasive sampling methods, and high-sensitivity sensing techniques are essential for comprehensively monitoring a diverse range of biomarkers. Additionally, a detailed understanding and

characterization of the composition of each non-invasive biofluid and its relationship with blood biomarkers are critical for enhancing the acceptance of wearable technology in the healthcare domain. Furthermore, advancing integrated multimodal sensors that feature drug feedback control for closed-loop therapeutics presents a promising direction for future developments. Such sophisticated devices have the potential to continuously gather vital parameters in real-time while adapting treatment plans based on immediate feedback, thereby transforming conventional static treatments into dynamic, adaptive interventions.

Received 12 October 2024; accepted 31 December 2024;

published online 14 February 2025

- Karimi-Maleh H, Orooji Y, Karimi F, *et al.* A critical review on the use of potentiometric based biosensors for biomarkers detection. *Biosens Bioelectron*, 2021, 184: 113252
- Pataranutaporn P, Jain A, Johnson CM, *et al.* Wearable lab on body: combining sensing of biochemical and digital markers in a wearable device. In: 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). Berlin: IEEE, 2019. 3327–3332
- Li X, Zhang F, Liu Q. Portable and wearable sensing technologies for biochemical detection. In: Liu Q (ed.). *Portable and Wearable Sensing Systems: Techniques, Fabrication, and Biochemical Detection*. Wiley, 2024
- Nimse SB, Sonawane MD, Song KS, *et al.* Biomarker detection technologies and future directions. *Analyst*, 2016, 141: 740–755
- Erdem A, Eksin E, Senturk H, *et al.* Recent developments in wearable biosensors for healthcare and biomedical applications. *TrAC Trends Anal Chem*, 2024, 171: 117510
- Rao G, Srinivasan V, Sheffield Z, *et al.* Wearable transdermal biosensors. In: Mitsubayashi K (ed.). *Wearable Biosensing in Medicine and Healthcare*. Singapore: Springer, 2024. 89–109
- Alves T, Deroco P, Wachholz Junior D, *et al.* Wireless wearable electrochemical sensors: a review. *Braz J Anal Chem*, 2021, 8: 22–50
- Shanbhag MM, Manasa G, Mascarenhas RJ, *et al.* Fundamentals of bio-electrochemical sensing. *Chem Eng J Adv*, 2023, 16: 100516
- Mohan AMV, Rajendran V, Mishra RK, *et al.* Recent advances and perspectives in sweat based wearable electrochemical sensors. *TrAC Trends Anal Chem*, 2020, 131: 116024
- Sempionatto JR, Gao W. Wearable chemosensors. In: Morales-Narvaez E, Dincer C (eds.). *Wearable Physical, Chemical and Biological Sensors*. Elsevier, 2022. 219–234
- Pillai S, Upadhyay A, Sayson D, *et al.* Advances in medical wearable biosensors: design, fabrication and materials strategies in healthcare monitoring. *Molecules*, 2021, 27: 165
- Jung HH, Lee H, Yea J, *et al.* Wearable electrochemical sensors for real-time monitoring in diabetes mellitus and associated complications. *Soft Sci*, 2024, 4: 15
- Gao F, Liu C, Zhang L, *et al.* Wearable and flexible electrochemical sensors for sweat analysis: a review. *Microsyst Nanoeng*, 2023, 9: 1
- Qiao L, Benziger MR, Subramony JA, *et al.* Advances in sweat wearables: sample extraction, real-time biosensing, and flexible platforms. *ACS Appl Mater Interfaces*, 2020, 12: 34337–34361
- Park H, Park W, Lee CH. Electrochemically active materials and wearable biosensors for the *in situ* analysis of body fluids for human healthcare. *NPG Asia Mater*, 2021, 13: 23
- Arakawa T, Dao DV, Mitsubayashi K. Biosensors and chemical sensors for healthcare monitoring: a review. *IEEE Trans Elec Engng*, 2022, 17: 626–636
- Sharma R, Geranpayehvaghei M, Ejeian F, *et al.* Recent advances in polymeric nanostructured ion selective membranes for biomedical applications. *Talanta*, 2021, 235: 122815
- Huang X, Li H, Li J, *et al.* Transient, implantable, ultrathin biofuel cells enabled by laser-induced graphene and gold nanoparticles composite. *Nano Lett*, 2022, 22: 3447–3456
- Huang X, Liu Y, Park W, *et al.* Stretchable magnesium–air battery based on dual ions conducting hydrogel for intelligent biomedical applications. *InfoMat*, 2022, 5: e12388
- Antony A. Flexible and wearable biosensors: revolutionizing health monitoring. In: Mathew R, Ajayan J (eds.). *Biosensors: Developments, Challenges and Perspectives*. Singapore: Springer, 2024. 237–258
- Li J, Zhou Y, Jiang C, *et al.* Recent advances in passive cooling materials for thermal management in flexible electronics. *J Mater Chem C*, 2024, 12: 12179–12206
- Kim J, Campbell AS, de Ávila BEF, *et al.* Wearable biosensors for healthcare monitoring. *Nat Biotechnol*, 2019, 37: 389–406
- Sempionatto JR, Lasalde-Ramirez JA, Mahato K, *et al.* Wearable chemical sensors for biomarker discovery in the omics era. *Nat Rev Chem*, 2022, 6: 899–915
- Kulkarni MB, Rajagopal S, Prieto-Simón B, *et al.* Recent advances in smart wearable sensors for continuous human health monitoring. *Talanta*, 2024, 272: 125817
- Bakri MH, Özarslan AC, Erarslan A, *et al.* Biomedical applications of wearable biosensors. *Next Mater*, 2024, 3: 100084
- Song Z, Zhou S, Qin Y, *et al.* Flexible and wearable biosensors for monitoring health conditions. *Biosensors*, 2023, 13: 630
- Liu T, Liu L, Gou G, *et al.* Recent advancements in physiological, biochemical, and multimodal sensors based on flexible substrates: strategies, technologies, and integrations. *ACS Appl Mater Interfaces*, 2023, 15: 21721–21745
- Shi Z, Dai C, Fang F, *et al.* Wearable sensors for *in situ* biofluid analysis. In: Liu Q (ed.). *Portable and Wearable Sensing Systems: Techniques, Fabrication, and Biochemical Detection*. Wiley, 2024. 139–156
- Keum DH, Kim SK, Koo J, *et al.* Wireless smart contact lens for diabetic diagnosis and therapy. *Sci Adv*, 2020, 6: eaba3252
- Patle S, Rotake D. Recent advances, technological challenges and requirements to predict the future trends in wearable sweat sensors: a critical review. *MicroChem J*, 2024, 200: 110457
- Yu Y, Nyein HYY, Gao W, *et al.* Flexible electrochemical bioelectronics: the rise of *in situ* bioanalysis. *Adv Mater*, 2020, 32: e1902083
- Bandodkar AJ, Jeerapan I, Wang J. Wearable chemical sensors: present challenges and future prospects. *ACS Sens*, 2016, 1: 464–482
- Schazmann B, Morris D, Slater C, *et al.* A wearable electrochemical sensor for the real-time measurement of sweat sodium concentration. *Anal Methods*, 2010, 2: 342–348
- Yeung KK, Huang T, Hua Y, *et al.* Recent advances in electrochemical sensors for wearable sweat monitoring: a review. *IEEE Sens J*, 2021, 21: 14522–14539
- Kim SK, Lee GH, Jeon C, *et al.* Bimetallic nanocatalysts immobilized in nanoporous hydrogels for long-term robust continuous glucose monitoring of smart contact lens. *Adv Mater*, 2022, 34: 2110536
- Zhao G, Li Z, Huang X, *et al.* Recent advances of sweat sampling, sensing, energy-harvesting and data-display toward flexible sweat electronics. *Soft Sci*, 2024, 4:
- Liu C, Xu T, Wang D, *et al.* The role of sampling in wearable sweat sensors. *Talanta*, 2020, 212: 120801
- Gao W, Emaminejad S, Nyein HYY, *et al.* Fully integrated wearable sensor arrays for multiplexed *in situ* perspiration analysis. *Nature*, 2016, 529: 509–514
- Dutkiewicz EP, Lin JD, Tseng TW, *et al.* Hydrogel micropatches for sampling and profiling skin metabolites. *Anal Chem*, 2014, 86: 2337–2344
- Choi J, Kang D, Han S, *et al.* Thin, soft, skin-mounted microfluidic networks with capillary bursting valves for chrono-sampling of sweat. *Adv Healthcare Mater*, 2017, 6: 1601355
- Ita K. Recent progress in transdermal sonophoresis. *Pharm Dev Tech*, 2017, 22: 458–466
- Xu C, Jiang D, Ge Y, *et al.* A PEDOT:PSS conductive hydrogel incorporated with Prussian blue nanoparticles for wearable and non-invasive monitoring of glucose. *Chem Eng J*, 2022, 431: 134109
- Ray TR, Choi J, Bandodkar AJ, *et al.* Bio-integrated wearable systems: a comprehensive review. *Chem Rev*, 2019, 119: 5461–5533

- 44 De la Paz E, Saha T, Del Caño R, *et al.* Non-invasive monitoring of interstitial fluid lactate through an epidermal iontophoretic device. *Talanta*, 2023, 254: 124122
- 45 Xu G, Huang X, Shi R, *et al.* Triboelectric nanogenerator enabled sweat extraction and power activation for sweat monitoring. *Adv Funct Mater*, 2024, 34: 2310777
- 46 Kim J, Sempionatto JR, Imani S, *et al.* Simultaneous monitoring of sweat and interstitial fluid using a single wearable biosensor platform. *Adv Sci*, 2018, 5: 1800880
- 47 Yang J, Gong X, Chen S, *et al.* Development of smartphone-controlled and microneedle-based wearable continuous glucose monitoring system for home-care diabetes management. *ACS Sens*, 2023, 8: 1241–1251
- 48 Mohan AMV, Windmiller JR, Mishra RK, *et al.* Continuous minimally-invasive alcohol monitoring using microneedle sensor arrays. *Biosens Bioelectron*, 2017, 91: 574–579
- 49 Lin PH, Sheu SC, Chen CW, *et al.* Wearable hydrogel patch with noninvasive, electrochemical glucose sensor for natural sweat detection. *Talanta*, 2022, 241: 123187
- 50 Saha T, Songkakul T, Knisely CT, *et al.* Wireless wearable electrochemical sensing platform with zero-power osmotic sweat extraction for continuous lactate monitoring. *ACS Sens*, 2022, 7: 2037–2048
- 51 Al Sulaiman D, Chang JYH, Bennett NR, *et al.* Hydrogel-coated microneedle arrays for minimally invasive sampling and sensing of specific circulating nucleic acids from skin interstitial fluid. *ACS Nano*, 2019, 13: 9620–9628
- 52 Xiao J, Zhang S, Liu Q, *et al.* Microfluidic-based plasmonic microneedle biosensor for uric acid ultrasensitive monitoring. *Sens Actuat B-Chem*, 2024, 398: 134685
- 53 Abbasiasl T, Mirlou F, Mirzajani H, *et al.* A wearable touch-activated device integrated with hollow microneedles for continuous sampling and sensing of dermal interstitial fluid. *Adv Mater*, 2024, 36: e2304704
- 54 Yang Q, Rosati G, Abarintos V, *et al.* Wearable and fully printed microfluidic nanosensor for sweat rate, conductivity, and copper detection with healthcare applications. *Biosens Bioelectron*, 2022, 202: 114005
- 55 Hu Y, Wang L, Li J, *et al.* Thin, soft, skin-integrated electronics for real-time and wireless detection of uric acid in sweat. *Int J Smart Nano Mater*, 2023, 14: 406–419
- 56 Huang X, Li J, Liu Y, *et al.* Epidermal self-powered sweat sensors for glucose and lactate monitoring. *Bio-des Manuf*, 2021, 5: 201–209
- 57 Bandodkar AJ, Jeang WJ, Ghaffari R, *et al.* Wearable sensors for biochemical sweat analysis. *Annu Rev Anal Chem*, 2019, 12: 1–22
- 58 Faham S, Salimi A, Ghavami R. Electrochemical-based remote biomarker monitoring: toward Internet of Wearable Things in telemedicine. *Talanta*, 2023, 253: 123892
- 59 Shu Y, Su T, Lu Q, *et al.* Highly stretchable wearable electrochemical sensor based on Ni-Co MOF nanosheet-decorated Ag/rGO/PU fiber for continuous sweat glucose detection. *Anal Chem*, 2021, 93: 16222–16230
- 60 Mathew R, Ajayan J. Bioresorbable biosensors. In: Mathew R, Ajayan J (eds.). *Biosensors: Developments, Challenges and Perspectives*. Singapore: Springer, 2024. 13–25
- 61 Li G, Huang K, Deng J, *et al.* Highly conducting and stretchable double-network hydrogel for soft bioelectronics. *Adv Mater*, 2022, 34: 2200261
- 62 Sempionatto JR, Brazaca LC, García-Carmona L, *et al.* Eyeglasses-based tear biosensing system: non-invasive detection of alcohol, vitamins and glucose. *Biosens Bioelectron*, 2019, 137: 161–170
- 63 Jin X, Li G, Xu T, *et al.* Fully integrated flexible biosensor for wearable continuous glucose monitoring. *Biosens Bioelectron*, 2022, 196: 113760
- 64 Nyein HYY, Bariya M, Kivimäki L, *et al.* Regional and correlative sweat analysis using high-throughput microfluidic sensing patches toward decoding sweat. *Sci Adv*, 2019, 5: eaaw9906
- 65 Yang Y, Song Y, Bo X, *et al.* A laser-engraved wearable sensor for sensitive detection of uric acid and tyrosine in sweat. *Nat Biotechnol*, 2020, 38: 217–224
- 66 Tran VV, Lee S, Lee D, *et al.* Recent developments and implementations of conductive polymer-based flexible devices in sensing applications. *Polymers*, 2022, 14: 3730
- 67 Thirumalai D, Santhamoorthy M, Kim SC, *et al.* Conductive polymer-based hydrogels for wearable electrochemical biosensors. *Gels*, 2024, 10: 459
- 68 Zheng Z, Zhu R, Peng I, *et al.* Wearable and implantable biosensors: mechanisms and applications in closed-loop therapeutic systems. *J Mater Chem B*, 2024, 12: 8577–8604
- 69 Liu G, Lv Z, Batool S, *et al.* Biocompatible material-based flexible biosensors: from materials design to wearable/implantable devices and integrated sensing systems. *Small*, 2023, 19: e2207879
- 70 Chen S, Qi J, Fan S, *et al.* Flexible wearable sensors for cardiovascular health monitoring. *Adv Healthcare Mater*, 2021, 10: 2100116
- 71 Zhang B, Li J, Zhou J, *et al.* A three-dimensional liquid diode for soft, integrated permeable electronics. *Nature*, 2024, 628: 84–92
- 72 Drexelius A, Yuan Y, Friedel M, *et al.* Wearable biosensors. In: Morales-Narvaez E, Dincer C (eds.). *Wearable Physical, Chemical and Biological Sensors*. Elsevier, 2022. 235–254
- 73 Lahcen AA, Amine A. Biorecognition elements. In: Morales-Narvaez E, Dincer C (eds.). *Wearable Physical, Chemical and Biological Sensors*. Elsevier, 2022. 41–70
- 74 Schachinger F, Chang H, Scheiblbrandner S, *et al.* Amperometric biosensors based on direct electron transfer enzymes. *Molecules*, 2021, 26: 4525
- 75 Kilic NM, Singh S, Keles G, *et al.* Novel approaches to enzyme-based electrochemical nanobiosensors. *Biosensors*, 2023, 13: 622
- 76 Cheng X, Wang B, Zhao Y, *et al.* A mediator-free electroenzymatic sensing methodology to mitigate ionic and electroactive interferents' effects for reliable wearable metabolite and nutrient monitoring. *Adv Funct Mater*, 2020, 30: 1908507
- 77 Kalasin S, Sangnuang P, Khownarumit P, *et al.* Salivary creatinine detection using a Cu(I)/Cu(II) catalyst layer of a supercapacitive hybrid sensor: a wireless IoT device to monitor kidney diseases for remote medical mobility. *ACS BioMater Sci Eng*, 2020, 6: 5895–5910
- 78 Li G, Wang C, Chen Y, *et al.* Dual structural design of platinum-nickel hydrogels for wearable glucose biosensing with ultrahigh stability. *Small*, 2023, 19: e2206868
- 79 Naresh V, Lee N. A review on biosensors and recent development of nanostructured materials-enabled biosensors. *Sensors*, 2021, 21: 1109
- 80 Morales MA, Halpern JM. Guide to selecting a biorecognition element for biosensors. *Bioconjugate Chem*, 2018, 29: 3231–3239
- 81 de Araújo NS, Moreira AS, Abreu RS, *et al.* Aptamer-based recognition of breast tumor cells: a new era for breast cancer diagnosis. *Int J Mol Sci*, 2024, 25: 840
- 82 Gui R, Jin H, Guo H, *et al.* Recent advances and future prospects in molecularly imprinted polymers-based electrochemical biosensors. *Biosens Bioelectron*, 2018, 100: 56–70
- 83 Herrera-Chacón A, Cetó X, del Valle M. Molecularly imprinted polymers—towards electrochemical sensors and electronic tongues. *Anal Bioanal Chem*, 2021, 413: 6117–6140
- 84 Takaloo S, Moghimi Zand M. Wearable electrochemical flexible biosensors: with the focus on affinity biosensors. *Sens Bio-Sens Res*, 2021, 32: 100403
- 85 Taheri N, Khoshshafar H, Ghanei M, *et al.* Dual-template rectangular nanotube molecularly imprinted polypyrrole for label-free impedimetric sensing of AFP and CEA as lung cancer biomarkers. *Talanta*, 2022, 239: 123146
- 86 Ma Y, Shen XL, Wang HS, *et al.* MIPs-graphene nanoplatelets-MWCNTs modified glassy carbon electrode for the determination of cardiac troponin I. *Anal Biochem*, 2017, 520: 9–15
- 87 Li G, Wen D. Wearable biochemical sensors for human health monitoring: sensing materials and manufacturing technologies. *J Mater Chem B*, 2020, 8: 3423–3436
- 88 Zhao K, Kang B, Zhou B. Wearable electrochemical sensors for monitoring of inorganic ions and pH in sweat. *Int J Electrochem Sci*, 2022, 17: 220452
- 89 Benoudjit AM, Shohibuddin IUS, Bader MM, *et al.* Components of all-solid-state ion-selective electrodes (AS-ISEs). In: Siddiquee S, Hong MGJ, Rahman MM (eds.). *Composite Materials: Applications in*

- Engineering, Biomedicine and Food Science. Cham: Springer, 2020. 351–366
- 90 Torrente-Rodríguez RM, Tu J, Yang Y, *et al.* Investigation of cortisol dynamics in human sweat using a graphene-based wireless mHealth system. *Matter*, 2020, 2: 921–937
 - 91 Polidori G, Tonello S, Serpelloni M. Ion-selective all-solid-state printed sensors: a systematic review. *IEEE Sens J*, 2024, 24: 7375–7394
 - 92 Kusumkar VV, Galamboš M, Vigišová E, *et al.* Ion-imprinted polymers: synthesis, characterization, and adsorption of radionuclides. *Materials*, 2021, 14: 1083
 - 93 Darroudi A. Electrophoretic deposition of graphene oxide on screen-imprinted carbon electrode and its modification using Ni²⁺-imprinted polymer as ionophore by a potentiometric sensor for determination of nickel ions. *Anal Bioanal Chem Res*, 2023, 10: 63–70
 - 94 Shao Y, Ying Y, Ping J. Recent advances in solid-contact ion-selective electrodes: functional materials, transduction mechanisms, and development trends. *Chem Soc Rev*, 2020, 49: 4405–4465
 - 95 Sempionatto JR, Jeerapan I, Krishnan S, *et al.* Wearable chemical sensors: emerging systems for on-body analytical chemistry. *Anal Chem*, 2020, 92: 378–396
 - 96 Min J, Sempionatto JR, Teymourian H, *et al.* Wearable electrochemical biosensors in North America. *Biosens Bioelectron*, 2021, 172: 112750
 - 97 Huang X, Liu Y, Zhou J, *et al.* Garment embedded sweat-activated batteries in wearable electronics for continuous sweat monitoring. *npj Flex Electron*, 2022, 6: 10
 - 98 Zhao J, Lin Y, Wu J, *et al.* A fully integrated and self-powered smartwatch for continuous sweat glucose monitoring. *ACS Sens*, 2019, 4: 1925–1933
 - 99 Bollella P, Sharma S, Cass AEG, *et al.* Microneedle-based biosensor for minimally-invasive lactate detection. *Biosens Bioelectron*, 2019, 123: 152–159
 - 100 Fariha R, Jabrah M, Hill C, *et al.* Simultaneous detection of salivary cortisol and cortisone using an automated high-throughput sample preparation method for LC-MS/MS. *SLAS Tech*, 2022, 27: 237–246
 - 101 Dai B, Zhou R, Ping J, *et al.* Recent advances in carbon nanotube-based biosensors for biomolecular detection. *TrAC Trends Anal Chem*, 2022, 154: 116658
 - 102 Ciui B, Martin A, Mishra RK, *et al.* Wearable wireless tyrosinase bandage and microneedle sensors: toward melanoma screening. *Adv Healthcare Mater*, 2018, 7: 1701264
 - 103 Anbuselvam B, Gunasekaran BM, Srinivasan S, *et al.* Wearable biosensors in cardiovascular disease. *Clinica Chim Acta*, 2024, 561: 119766
 - 104 Sempionatto JR, Lin M, Yin L, *et al.* An epidermal patch for the simultaneous monitoring of haemodynamic and metabolic biomarkers. *Nat Biomed Eng*, 2021, 5: 737–748
 - 105 García-Carmona L, Martín A, Sempionatto JR, *et al.* Pacifier biosensor: toward noninvasive saliva biomarker monitoring. *Anal Chem*, 2019, 91: 13883–13891
 - 106 Arakawa T, Tomoto K, Nitta H, *et al.* A wearable cellulose acetate-coated mouthguard biosensor for *in vivo* salivary glucose measurement. *Anal Chem*, 2020, 92: 12201–12207
 - 107 Romanholo PVV, Razzino CA, Raymundo-Pereira PA, *et al.* Biomimetic electrochemical sensors: new horizons and challenges in biosensing applications. *Biosens Bioelectron*, 2021, 185: 113242
 - 108 Chen Y, Li G, Mu W, *et al.* Nonenzymatic sweat wearable uric acid sensor based on N-doped reduced graphene oxide/Au dual aerogels. *Anal Chem*, 2023, 95: 3864–3872
 - 109 Toi PT, Trung TQ, Dang TML, *et al.* Highly electrocatalytic, durable, and stretchable nanohybrid fiber for on-body sweat glucose detection. *ACS Appl Mater Interfaces*, 2019, 11: 10707–10717
 - 110 Jang J, Kim J, Shin H, *et al.* Smart contact lens and transparent heat patch for remote monitoring and therapy of chronic ocular surface inflammation using mobiles. *Sci Adv*, 2021, 7: eabf7194
 - 111 Cheng C, Li X, Xu G, *et al.* Battery-free, wireless, and flexible electrochemical patch for *in situ* analysis of sweat cortisol via near field communication. *Biosens Bioelectron*, 2021, 172: 112782
 - 112 Wang B, Zhao C, Wang Z, *et al.* Wearable aptamer-field-effect transistor sensing system for noninvasive cortisol monitoring. *Sci Adv*, 2022, 8: eabk0967
 - 113 Singh NK, Chung S, Chang AY, *et al.* A non-invasive wearable stress patch for real-time cortisol monitoring using a pseudoknot-assisted aptamer. *Biosens Bioelectron*, 2023, 227: 115097
 - 114 Wang M, Yang Y, Min J, *et al.* A wearable electrochemical biosensor for the monitoring of metabolites and nutrients. *Nat Biomed Eng*, 2022, 6: 1225–1235
 - 115 Zhang Q, Jiang D, Xu C, *et al.* Wearable electrochemical biosensor based on molecularly imprinted Ag nanowires for noninvasive monitoring lactate in human sweat. *Sens Actuat B-Chem*, 2020, 320: 128325
 - 116 Mei X, Yang J, Yu X, *et al.* Wearable molecularly imprinted electrochemical sensor with integrated nanofiber-based microfluidic chip for *in situ* monitoring of cortisol in sweat. *Sens Actuat B-Chem*, 2023, 381: 133451
 - 117 Wardak C, Pietrzak K, Morawska K, *et al.* Ion-selective electrodes with solid contact based on composite materials: a review. *Sensors*, 2023, 23: 5839
 - 118 Jadon N, Hosseinzadeh B, Kaya SI, *et al.* Emerging trends of ion-selective electrodes in pharmaceutical applications. *Electrochim Acta*, 2024, 488: 144204
 - 119 Zdrachek E, Bakker E. Potentiometric sensing. *Anal Chem*, 2021, 93: 72–102
 - 120 Lyu Y, Gan S, Bao Y, *et al.* Solid-contact ion-selective electrodes: response mechanisms, transducer materials and wearable sensors. *Membranes*, 2020, 10: 128
 - 121 Tamara MR, Lelono D, Roto R, *et al.* All-solid-state astringent taste sensor using polypyrrole-carbon black composite as ion-electron transducer. *Sens Actuat A-Phys*, 2023, 351: 114170
 - 122 Yoon JH, Kim SM, Eom Y, *et al.* Extremely fast self-healable bio-based supramolecular polymer for wearable real-time sweat-monitoring sensor. *ACS Appl Mater Interfaces*, 2019, 11: 46165–46175
 - 123 Xu J, Zhang Z, Gan S, *et al.* Highly stretchable fiber-based potentiometric ion sensors for multichannel real-time analysis of human sweat. *ACS Sens*, 2020, 5: 2834–2842
 - 124 Parrilla M, Cuartero M, Crespo GA. Wearable potentiometric ion sensors. *TrAC Trends Anal Chem*, 2019, 110: 303–320
 - 125 Sempionatto JR, Martin A, García-Carmona L, *et al.* Skin-worn soft microfluidic potentiometric detection system. *Electroanalysis*, 2019, 31: 239–245
 - 126 Liu H, Gu Z, Zhao Q, *et al.* Printed circuit board integrated wearable ion-selective electrode with potential treatment for highly repeatable sweat monitoring. *Sens Actuat B-Chem*, 2022, 355: 131102
 - 127 Li J, Ling M, Shui J, *et al.* Smart grazing in Tibetan Plateau: development of a ground-air-space integrated low-cost Internet of Things system for yak monitoring. *Wireless Commun Mobile Computing*, 2022, 2022: 1–18
 - 128 Zhai Q, Yap LW, Wang R, *et al.* Vertically aligned gold nanowires as stretchable and wearable epidermal ion-selective electrode for non-invasive multiplexed sweat analysis. *Anal Chem*, 2020, 92: 4647–4655
 - 129 Parrilla M, Ortiz-Gómez I, Cánovas R, *et al.* Wearable potentiometric ion patch for on-body electrolyte monitoring in sweat: toward a validation strategy to ensure physiological relevance. *Anal Chem*, 2019, 91: 8644–8651
 - 130 An Q, Gan S, Xu J, *et al.* A multichannel electrochemical all-solid-state wearable potentiometric sensor for real-time sweat ion monitoring. *Electrochem Commun*, 2019, 107: 106553
 - 131 Soliman RM, Rostom Y, Mahmoud AM, *et al.* Novel fabricated potentiometric sensors for selective determination of carboxamine with different greenness evaluation perspectives. *MicroChem J*, 2023, 187: 108381
 - 132 Zhizhin KY, Turyshev ES, Kopytin AV, *et al.* Polymer nanocarbon materials as ion-to-electron transducers in solid-contact ion-selective electrodes. *NanoSyst-Phys Chem Math*, 2022, 13: 688–697
 - 133 Ozer T, Agir I, Henry CS. Low-cost Internet of Things (IoT)-enabled a wireless wearable device for detecting potassium ions at the point of care. *Sens Actuat B: Chem*, 2022, 365: 131961
 - 134 Zou J, Lin Z, Yuan Z, *et al.* High-strength, high-toughness regenerated cellulose/graphene oxide nanofluidic membrane with highly oriented

and charged nanochannels for wearable sweat-monitoring systems. *Chem Eng J*, 2023, 467: 143485

Acknowledgement This work was supported by the National Key Research and Development Program of China (2022YFB3205600), Research Grants Council of the Hong Kong Special Administrative Region (RFS2324-1S03), National Natural Science Foundation of China (62122002), The City University of Hong Kong (9667221, 9667246, 9680322, and 9667199), and in part of the InnoHK Project on Project 2.2—AI-based 3D ultrasound imaging algorithm at Hong Kong Centre for Cerebro-Cardiovascular Health Engineering (COCHE).

Funding note Open access publishing enabled by City University of Hong Kong Library's agreement with Springer Nature.

Author contributions Jiao Y conceived the ideas, conducted the literature review and wrote the manuscript. Yu X did the review and editing, and supervision.

Conflict of interest The authors declare that they have no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.



Yanli Jiao is currently a PhD student at the College of Engineering/Department of Biomedical Engineering at the City University of Hong Kong. She received her Master's degree in food process engineering from the Illinois Institute of Technology. She has an interdisciplinary background in chemistry, biomedical engineering and microfabrication. Her research is mainly focused on flexible nanomaterials, wearable bioelectronics for metabolic monitoring, and analytical chemistry.



Xinge Yu is currently a Member of the Hong Kong Young Academy of Sciences, and Associate Director for Institute of Digital Medicine, Hong Kong Centre for Cerebro-cardiovascular Health Engineering. He is the recipient of RGC Research Fellow, Innovators under 35 China (MIT Technology Review), NSFC Excellent Young Scientist Grant (Hong Kong & Macao), New Innovator of IEEE NanoMed, MINE Young Scientist Award, Gold Medal in the Inventions Geneva, CityU Outstanding Research Award, and Stanford's top 2% most highly cited scientists etc. His research group focuses on skin-integrated electronics and systems for VR and biomedical applications.

可穿戴电化学传感器在生化标志物原位检测领域的最新研究进展

焦艳丽^{1,2}, 于欣格^{1,2*}

摘要 近年来, 可穿戴电化学传感器在生化分析领域得到了广泛应用. 这些传感器通过将柔性电极和敏感识别元件集成于柔性衬底上, 实现了对生物流体中目标生化分子的非侵入性、原位、实时及连续监测, 并保持了高选择性和灵敏度. 本文综述了近年来用于生化标志物检测的可穿戴电化学传感器的核心元件选择原则及其最新进展. 首先, 我们概述了设计用于检测生物流体中生物标志物的可穿戴传感器的基本考虑因素, 包括采样技术、材料选择、设计参数、识别元素、传感策略、功率需求、数据处理及传感器集成. 生物技术和材料科学的进步显著提升了识别元件的效率, 从而在复杂基质中实现了对目标成分的选择性和灵敏检测. 此外, 纳米材料与导电聚合物的结合显著增强了柔性电子器件的传感能力. 随后, 我们探讨了可穿戴电化学传感器对生化标志物进行原位检测的最新进展. 这些传感器利用先进材料、优化机械结构及采用多种传导机制制备来实现对不同生物标志物的筛选和检测. 这些技术创新不仅提高了用户舒适度, 还在灵敏度、可靠性和监测能力方面取得了显著进步. 最后, 我们讨论了当前面临的挑战及未来前景, 旨在实现临床导向的可穿戴电化学传感器, 以支持疾病监测和个性化医疗.